



A physics-informed neural network surrogate model and many-objective optimization algorithm for coupled multi-energy systems in smart grids

Jingbo Zhang^{a,b}, Xingjuan Cai^{b,*}, Zhihua Cui^{b,*}, Jinjun Chen^c

^a Shanxi Key Laboratory of Advanced Control and Equipment intelligence, Taiyuan University of Science and Technology, Taiyuan, 030024, Shanxi, China

^b Shanxi Key Laboratory of Big Data Analysis and Parallel Computing, Taiyuan University of Science and Technology, Taiyuan, 030024, Shanxi, China

^c Department of Computing Technologies, Swinburne University of Technology, Melbourne, Victoria, Australia

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ABSTRACT

The scope of smart grids is progressively extending toward Integrated Energy Systems (IES) that couple electricity with gas, heating, and cooling. Due to the unsteady-state physical characteristics inherent in the transmission of gas, heat, and cooling resources, IES scheduling must not only balance multiple typical objectives but also account for the dynamic coupling of heterogeneous physical domains. To address these challenges, this paper formulates a Many-objective Optimization Model for Coupled Multi-Energy Flows (MaOCMFM) with partial differential equations (PDEs) in IES, which captures the dynamic physical behaviors of electricity, gas, heat, and cooling subsystems. Building upon this model, we propose a Probabilistic Contributing Many-objective Evolutionary Algorithm enhanced by a Physics-Informed Neural Network surrogate model (PC-MaOEA-PINN). Cubic B-spline functions are employed to achieve a continuous representation of the decision variables, while multi-physics constraints are embedded into the loss function of the surrogate model. This design enables efficient approximation of the objective function with a limited number of samples and facilitates focused exploration in critical evolutionary regions, thereby accelerating population convergence. The effectiveness of the proposed model and algorithm is validated on 9 typical scheduling days across four simulated IES scenarios.

1. Introduction

With the continued advancement of the Dual Carbon goals and the deep transformation of the energy structure, Integrated Energy Systems (IES), which feature multi-energy complementarity and source-load coordination, are increasingly becoming a critical infrastructure for the development of modern power systems. By enabling the coupling and coordinated optimization of electricity, gas, heating, and cooling, IES significantly enhance overall energy utilization efficiency and operational flexibility, offering new pathways for the evolution of smart grids (Fang et al., 2024; Kou et al., 2025).

Traditional power system scheduling problems are typically formulated as nonlinear optimization tasks subject to power flow balance, nodal power limits, and other constraints. With the integration of diverse energy carriers-particularly renewable energy sources (RES) and such as wind and photovoltaic (PV) power-IES aim to improve coordination efficiency across multi-energy domains (Atawi et al., 2024). However, the inclusion of such heterogeneous energy sources substantially alters the control variables, state variables, and constraint sets, thereby

intensifying the nonlinearity and complexity of the scheduling problem (Li et al., 2024).

Moreover, physical phenomena such as the compressibility in gas transport, inertial responses in heat systems, and strong nonlinear fluctuations in cooling demand induce non-steady-state behaviors during energy transmission. To achieve effective coupling and coordination among different energy carriers, dynamic modeling via partial differential equations (PDEs) becomes essential-posing a major challenge in IES optimization.

Current research efforts generally follow two main directions: (i) simplifying non-steady-state dynamics into quasi-steady or linearized models to enable tractable formulations under conventional power system optimization frameworks. This includes techniques such as equivalent load modeling, linear approximations of heat flows, and steady-state assumptions for gas dynamics, solved via nonlinear programming or metaheuristic optimization algorithms; and (ii) retaining the physical fidelity of system dynamics through PDE-based modeling, aiming to construct more realistic coupled optimization frameworks. The resulting PDEs are typically solved using numerical discretization methods

* Corresponding authors.

E-mail addresses: zhangjingbo@stu.tyust.edu.cn (J. Zhang), caixingjuan@tyust.edu.cn (X. Cai), cuizhihua@tyust.edu.cn (Z. Cui), jchen@swin.edu.au (J. Chen).

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such as finite difference, finite volume, finite element, or higher-order Galerkin methods.

In line with model simplification, Fang et al. (2017) developed a dynamic optimal energy flow model for gas-electric coupled systems by integrating transient gas flow with steady-state power flow and simplifying it into a single-stage linear programming problem. Zhang et al. (2023) proposed a piecewise-linear one-dimensional approximation for modeling the dynamic behaviors of heating and gas subsystems, formulating the joint transmission and distribution optimization task as a mixed-integer programming problem. Similarly, Li et al. (2021) employed a multilayer perceptron to construct a stochastic robust optimization model for community-level IES, incorporating horizontal and vertical complementarity strategies across electricity, gas, heating, and cooling. Huang et al. (2024) focused on the integration of battery energy storage systems (BESS) into IES planning, specifically addressing capacity sizing and degradation. Chen et al. (2023) built a nonlinear programming model that considers cogeneration, carbon capture, and power-to-gas operations, solved via interior-point methods. Yan et al. (2023) proposed a two-stage nonlinear programming approach to minimize investment, operational cost, carbon emissions, and RES curtailment losses. Li et al. (2024) addressed uncertainty in renewable generation via a distributed multi-objective robust optimization framework, where uncertainty is modeled using probabilistic sets and solved via column-and-constraint generation and the alternating direction method of multipliers (ADMM).

When nonlinear interdependencies exist among objectives, scalarization approaches become ineffective, highlighting the role of metaheuristic many-objective optimization algorithms. Guven et al. (2024) compared six metaheuristic algorithms for IES configuration optimization under integrated RES and BESS scenarios. Jia et al. (2024) employed the NSGA-II algorithm to solve a multi-objective IES optimization problem balancing economic benefits, carbon reduction, and system reliability. Liu et al. (2022) optimized IES capacities considering economic, environmental, and heat comfort criteria using multi-objective algorithms. Atawi et al. (2024) assessed the trade-off performance of multi-objective particle swarm optimization, African vulture optimization, and NSGA-II in wind-solar hybrid IES. Eriksson and Gray (2017) proposed a metaheuristic framework capable of handling up to four conflicting objectives, including capacity, availability, reliability, return on investment, and user costs. Zhu et al. (2025) introduced a multi-task evolutionary framework for IES multi-energy flow optimization, using three co-evolving populations to handle the main problem, relaxed constraints, and tightened constraints, thereby improving solution diversity and feasibility.

Although these methods have shown good practical performance, model simplifications inevitably introduce errors that may compromise physical consistency and scheduling reliability. In recent years, PDE-based modeling has gained attention for its ability to accurately capture the transport dynamics of media such as gas and heat. Yao et al. (2020) proposed a hybrid optimal energy flow model for IES, solving temperature PDE constraints via the finite difference method and optimizing temporal and spatial step sizes for unified scheduling. Liu et al. (2019) developed reasonable temporal-spatial discretization criteria for gas network energy flow optimization. Chen et al. (2020) introduced a fast decoupled method to accelerate energy flow computations. Zhang et al. (2024) reformulated the gas network PDEs in an integrated electricity-gas-heat system framework, significantly improving PDE solution accuracy.

Nonetheless, the high computational complexity of PDE-based multi-energy optimization remains a bottleneck. Surrogate modeling has emerged as a viable approach to mitigate this issue, especially when combined with evolutionary algorithms (Liu et al., 2024). Sun et al. (2025) proposed a preference-based surrogate-assisted multi-objective optimization algorithm to improve discrimination within the feasible region. Xie et al. (2023) introduced a two-level reward mechanism for model selection and infill criterion evaluation. Huang and Wang (2022)

utilized historical evolutionary data to construct and adaptively update surrogate models via binary space partitioning.

However, traditional surrogate models in evolutionary algorithms are primarily used to approximate expensive objective and constraint evaluations. Essentially, this constitutes a regression problem, where the main challenges lie in sampling the decision space effectively and selecting an appropriate model structure to generate useful data and labels—that is, the issue of filling criteria and model selection. Models such as Gaussian Processes, Kriging, and Support Vector Machines, however, struggle to approximate the numerical solutions of PDEs. This is because accurately capturing the complex, high-dimensional, and often nonlinear behavior of PDE solutions requires an impractically large amount of training data.

Physics-Informed Neural Networks (PINNs), as a novel paradigm for solving PDEs, explicitly incorporate governing equations into the loss function, thereby enforcing physical consistency. By leveraging this explicit physical information, PINNs significantly reduce the dependency on large datasets. While PINNs have been preliminarily applied in Integrated Energy Systems (IES) scenarios such as power flow analysis and state or parameter estimation, their systematic integration into multi-energy flow optimization remains largely unexplored.

It should be clarified that, although this work proposes a PINN-based surrogate model, its nature differs fundamentally from traditional surrogates used for expensive evaluation in evolutionary algorithms. Here, we borrow only the training procedure of surrogate models; we do not approximate objective or constraint values. Instead, the objective and constraint values are computed directly, and their evaluation is not considered computationally expensive.

To address the above challenges, this paper proposes a many-objective optimization modeling and evolutionary solving framework for IES with coupled multi-energy flows. The main contributions of this work are summarized as follows:

- (i) A Many-objective Optimization Model for Coupled Multi-Energy Flows (MaOCMF) is established in the form of partial differential equations (PDEs), providing a unified representation of the multi-physics coupling behaviors in IES. The model comprehensively considers typical optimization objectives such as economic cost, system profit, RES utilization rate, operational stability, and user comfort. It integrates multiple energy subsystems under the participation of RES and energy storage systems (ESS), including electrical, gas, heat, and cooling networks, and explicitly models their interactive coupling behaviors.
- (ii) To alleviate the computational cost associated with large-scale processing and PDE solving in MaOCMF, a Physics-Informed Neural Network Surrogate Model (PINNSM) based on cubic B-spline control points is developed. By embedding the multi-physics dynamic constraints into the loss function of the neural network, the proposed method enables physically consistent solutions with a limited number of training samples and enhances the accuracy of function approximation under sparse-data conditions.
- (iii) To address the efficiency challenge in the many-objective and high-dimensional decision spaces, a Probabilistic Contributing Many-objective Evolutionary Algorithm enhanced by PINNSM (PC-MaOEA-PINN) is designed. This algorithm leverages Shapley value-based variable importance analysis to identify key control variables and construct probabilistic evolution operators, thereby enabling adaptive search focusing and convergence acceleration in complex optimization landscapes.

The remainder of this paper is organized as follows: The motivation of this paper is given in Section 2; Section 3 introduces the proposed MaOCMF. Section 4 presents the design and implementation of the PC-MaOEA-PINN. Section 5 provides simulation studies and result analysis. Finally, Section 6 concludes the paper.

2. Motivation

The direct solution of PDEs constitutes the central contribution of this work. Previous approaches typically relied on discretization-based methods, such as finite difference, finite element, or finite volume methods, to approximate PDE solutions. While such methods are capable of producing high-fidelity solutions when sufficiently fine grids are employed, the associated computational burden becomes prohibitive.

In evolutionary algorithms, when evaluating objective or constraint violation values, it is often necessary to obtain PDE solutions at specific time steps or spatial locations (Tu et al., 2025). The conventional discretization-based approach requires iterative computation at each queried time or spatial point. In the context of IES scheduling problems, it is infeasible to pre-compute PDE solutions across all possible time steps and spatial locations, since although the governing PDEs remain the same, each individual corresponds to distinct boundary conditions. This renders grid-based methods computationally impractical (Lv et al., 2025).

Based on the above analysis, the solution of MaOCMF faces three main challenges: (i) The constraint contains PDE parameters, and solving the PDE equations typically requires high-precision numerical methods. However, evolutionary algorithms rely on comparing and selecting elite solutions from different individuals, which means that boundary condition changes require recalculating the PDE for each individual, resulting in significant computational cost; (ii) The convergence of individuals with temporal characteristics is particularly challenging, since random optimization alone often fails to capture the correct convergence direction in such cases (Zhang et al., 2021); (iii) In many-objective optimization spaces, balancing the trade-offs between multiple objectives is challenging, leading to insufficient selection pressure and, consequently, affecting both the solution quality and convergence (Li et al., 2025).

To address challenge (i), we adopt the concept of surrogate modeling in evolutionary algorithms and propose PINNSM to approximate PDE solutions. During the pre-training stage, we still leverage grid-based methods to generate data; however, thanks to the incorporation of explicit physical knowledge in PINNs, only a limited number of sampled points are required to approximate the PDE solution under the given boundary conditions. In other words, our method requires only a small number of grid iterations to extract representative interior points of the PDE, making it feasible to consider the PDE solution associated with each individual within the evolutionary optimization process.

The PCEO operator primarily addresses challenge (ii). To ensure continuous sampling of boundary conditions within PINNSM, we adopt spline functions to represent individuals, with their decision variables being the control points of the spline. This design inevitably introduces additional temporal features. Consequently, individuals acquire a temporal dimension, making traditional evolutionary strategies (e.g., GA, DE, or PSO), which generate new individuals through random variation or historical best positions, prone to leading the search toward incorrect convergence directions in our setting. To overcome this issue, PCEO calculates the Shapley values of each gene locus to quantify its contribution to the entire individual, while also assessing gradient variations. The operator then purposefully assigns different weights to loci with steep versus mild gradient changes, thereby guiding the search to focus on those loci with rapid variations, which increases the likelihood of generating high-quality individuals.

The CFNDS and CLPBI strategies are integrated as critical components of the environmental selection process. Their primary innovation lies in addressing the challenges posed by the temporal dimension. In particular, if an individual exhibits drastic changes at specific time instants, it may appear optimal over the entire scheduling horizon but is clearly impractical in real-world scenarios. To address this, we improved the non-dominated sorting procedure by introducing an integral-based relaxation mechanism. Furthermore, in CLPBI, elite individuals are selected through a combination of clustering strategies and the PBI in-

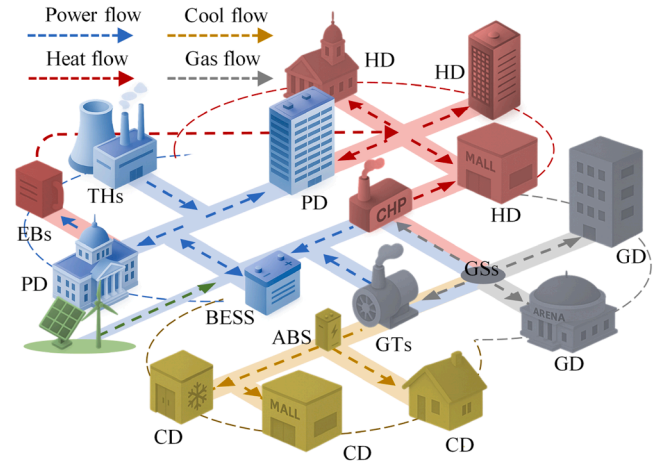


Fig. 1. Schematic diagram of the system architecture for the coupled multi-energy flow in IES.

dicator, which enhances the stability and robustness of the scheduling solutions.

3. Many-objective optimization model for coupled multi-energy flows

This paper focuses on a typical IES characterized by deep coupling of electrical, gas, heat, and cooling subsystems. The system architecture consists of an electrical subsystem, a gas subsystem, a heat subsystem, and a cooling subsystem. Within the electrical subsystem, power load (PQ) nodes are responsible not only for conventional electrical loads but also for supplying power to electric boilers (EBs) within the heat network for heating purposes. On the generation side, heat power units (THs), gas turbines (GTs), RES power units and combined heat and power units (CHPs) are included. The fuel for the gas turbines and CHPs is supplied by the gas network, which in turn receives natural gas from Gas Sources (GSs). In the heat subsystem, heat generation is provided by both EBs and CHPs. The cooling subsystem relies on absorption chillers (ABSs) as the primary cooling sources (CSs), which convert waste heat from gas turbine exhaust into cooling capacity. The overall system configuration is illustrated in Fig. 1.

In the coupled multi-energy flow IES, the subsystems do not operate independently but are tightly integrated through physical coupling and energy conversion mechanisms. A many-objective optimization problem is formulated, encompassing five representative objectives: economic cost, system profit, RES utilization rate, operational stability, and user comfort. This formulation aims to systematically investigate the synergistic coordination mechanisms and optimal operational strategies for multi-energy flow coupling within the IES, the notations involved are explained in Table 1. Furthermore, Tables A.1–A.6 present the IES system parameters employed in this study.

3.1. A coupled multi-energy flows system

In this subsection, we provide a detailed description of the configuration and constraints associated with the coupled multi-energy flows IES.

3.1.1. The power subsystem

It includes PV, PQ, and the slack node for reference. The generation side comprises THs, GTs, photovoltaic units (PHs), wind power units (WINS), and CHPs. These generation units are interconnected with load nodes through the power grid and must satisfy AC power flow constraints, as expressed in Eqs. (1)–(5).

Table 1
Explanation of notations in different subsystems.

Power		Heat & Cooling		Gas	
Notation	Explanation	Notation	Explanation	Notation	Explanation
N_T	Total IES nodes	N_{EB}	Number of EBs	N_{GAS}	Gas nodes
N_{ELE}	Power nodes	N_{CHP}	Number of CHPs	N_{GT}	GTs
N_{BESS}	Number of BESSs	$\eta_{CHP,GE}$	Gas-to-electricity eff.	η_{CHP}	Total eff. of CHPs
$N_{ELE,LOAD}$	Number of PD	$\eta_{CHP,GH}$	Gas-to-heat eff.	η_{GT}	Generation eff. of GTs
N_F	Number of THs	η_{ABS}	Conversion eff. of ABS	p	Fluid pressure
N_{EBR}	Power branches	η_{HRGS}	Conversion eff. of HRGS	ρ	Fluid density
N_{PV+WIN}	RES nodes	T_{fwd}	Supply temp.	u	Fluid velocity
η_{EB}	EB efficiency	T_{rev}	Return temp.	ρ_a	Reference density
η_{dis}	BESS discharge eff.	T_e	Node temp.	α	Pipe inclination
η_{ch}	BESS charge eff.	c_p	Specific heat	λ	Resistance coeff.
P_{ele}	Active power	m_h	Mass flow heating	d_g	Gas pipe diameter
Q_{ele}	Reactive power	u_{ch}	Charging BESS	m_g	Gas mass flow
θ	Phase angle	u_{dis}	Discharging BESS	Q_{LHV}	Gas lower heating value
V_m	Voltage amp.	c	Specific heat ratio		
I	Branch current	d_h	Pipe diameter		
r_{ij}, z_{ij}	Resistance/Reactance	T_{ea}	Env. temp.		
$D(j)$	Downstream nodes	λ_w	Pipe wall conductivity		
$U(j)$	Upstream nodes				
D_{cl}, U_{cl}	Power bounds				
β_{up}, β_{down}	Climbing cost				

Eq. (1) represents the standard AC branch power flow formulation, where P_{ij} denotes the real power flow from node i to node j , and denote the active and reactive power injections at node j , respectively. This expression captures the fundamental steady-state behavior of power transmission and forms the basis for multi-energy system optimization.

$$P_{ij} = \frac{V_i V_j}{X_{ij}} \cdot \sin(\theta_i - \theta_j) \quad (1)$$

$$P_j^{ele}(t) = \sum_{k \in \delta(j)} P_{jk}^{ele}(t) - \sum_{i \in \varphi(j)} (P_{ij}^{ele}(t) - I_{ij}^2(t) \cdot r_{ij}) \quad (2)$$

$$Q_j^{ele}(t) = \sum_{k \in \delta(j)} Q_{jk}^{ele}(t) - \sum_{i \in \varphi(j)} (Q_{ij}^{ele}(t) - I_{ij}^2(t) \cdot z_{ij}) \quad (3)$$

$$V_j^2(t) = V_i^2(t) - 2(r_{ij} \cdot P_{ij}(t) + z_{ij} \cdot Q_{ij}^{ele}(t)) + (r_{ij}^2 + x_{ij}^2) \cdot I_{ij}^2(t) \quad (4)$$

$$I_{ij}^2(t) = \frac{P_{ij}^2(t) + Q_{ij}^2(t)}{V_i^2(t)} \quad (5)$$

Unlike many existing approaches that linearize the nonlinear terms in AC power flow, we retain the quadratic components and evaluate the constraint violation degree (CVD) of each individual to determine its eligibility for the next iteration in the optimization process.

$$I_{ij}^{\min} \leq I_{ij} \leq I_{ij}^{\max} \quad (6)$$

$$V_j^{\min} \leq V_j \leq V_j^{\max} \quad (7)$$

$$P_i^{\min} \leq P_i^{ele}(t) \leq P_i^{\max} \quad (8)$$

$$Q_i^{\min} \leq Q_i^{ele}(t) \leq Q_i^{\max} \quad (9)$$

To ensure the operational reliability of the grid, each branch is further subject to voltage and power limit constraints, as given by Eqs. (6)–(9).

Here $P_i^{ele} = \{P_i^{TH}, P_i^{GT}, P_i^{PH}, P_i^{WIN}, P_i^{CHP,ele}, P_i^{BUY}, P_i^{BESS,dis}\}$ denotes the set of all generation nodes, TH, GT, WIN, power generated by the CHP, outsourced electric power, and discharge of power generated by BESS, Q_i^{ele} is similar. Notably, the energy storage system, when discharging, is modeled as a virtual generator and its output is incorporated into P_i^{ele} .

Additional operational constraints include ramping constraint and phase angle difference constraint. Such as, each dispatchable generation unit must comply with ramping limits on its power output, and the phase angle difference between adjacent buses must remain within a secure range to maintain system stability and prevent oscillatory behavior, as

Eqs. (10)–(11).

$$D_i^{cl} \leq \frac{dP_i^{ele}(t)}{dt} \leq U_i^{cl} \quad (10)$$

$$\frac{d\theta_i(t)}{dt} \leq \theta_{\max} \quad (11)$$

Finally, power balance constraints must be satisfied at each bus to ensure energy conservation, as shown in Eqs. (12)–(13), where $P_i^{ele,load}$ includes the load of EB and PQ node load. The matrix $G_{i,ele}$ represents the connection topology of the network, where each element takes a value of +1 or -1 to indicate power inflow or outflow.

$$\sum_{i=1}^{N_{ELE}} G_{i,ele} P_i^{ele}(t) + \sum_{i=1}^{N_{BESS}} u_i P_i^{bess}(t) = \sum_{i=1}^{N_{ELE,LOAD}} P_i^{ele,load}(t) \quad (12)$$

$$\sum_{i=1}^{N_{ELE}} G_{i,ele} Q_i^{ele}(t) + \sum_{i=1}^{N_{BESS}} u_i Q_i^{bess}(t) = \sum_{i=1}^{N_{ELE,LOAD}} Q_i^{ele,load}(t) \quad (13)$$

For the EB, the unit heat output $T_{e_i}^{eb,in}$ is related to the electric power load P_i^{EB} by the following equation:

$$P_i^{eb} = \sum_i^{N_{EB}} \left(\frac{T_{e_i}^{eb,in}}{\eta_{EB}} \right) \cdot c_p \cdot m_i^h(t) \quad (14)$$

The energy storage module is employed to enhance the accommodation capacity of RES in IES, and a BESS model is adopted to describe its operational behavior. The rate of change in battery energy capacity is expressed as shown in Eq. (15).

$$\frac{de_{bess}}{dt} = \frac{1}{E_{bess}^{\max}} \left(\eta_{ch} P_{bess}^{ch}(t) - \frac{P_{bess}^{dis}(t)}{\eta_{dis}} \right) \quad (15)$$

Here, $E_{bess}^{\max}$ denotes the maximum energy capacity of BESSs, P_{bess}^{dis} and P_{bess}^{ch} are the discharging and charging power of BESSs, respectively. The capacity constraints are given in Eq. (16).

$$E_{bess}^{\min} \leq \int_0^t \frac{de_{bess}}{dt} dt \leq E_{bess}^{\max} \quad (16)$$

In addition, to ensure the safe and controllable operation of BESS, its charging and discharging power must remain within prescribed limits, as defined in Eq. (17).

$$P_i^{bess \min} \leq P_i^{bess} \leq P_i^{bess \max} \quad (17)$$

Eq. (18) ensures that the BESS can operate in either the charging or discharging mode at any given time, but not both, thereby reflecting the physical limitations of practical systems.

$$P_i^{bess \min} \leq P_i^{bess} \leq P_i^{bess \max} \quad (18)$$

3.1.2. The gas subsystem

The GSs are represented by natural gas wells, while the gas load nodes (GDs) include not only regular end-use consumption but also the gas turbines used in the power subsystem, as well as combined CHPs serving both the power and heat subsystems. Gas is transported via a pipeline network. To accurately capture the transient gas flow dynamics in pipelines, a one-dimensional compressible fluid dynamics equation is employed, as formulated in Eq. (19).

$$\begin{cases} \frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial p}{\partial x} + g(\rho - \rho_a) \sin \alpha + \frac{\lambda}{d} \frac{\rho u^2}{2} = 0, \\ \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0 \end{cases} \quad (19)$$

This equation characterizes the coupled evolution of pressure, velocity, and density of the gas over time and space. In the modeling process, the initial conditions are predefined based on actual operating states, while Dirichlet boundary conditions are imposed and parameterized through decision variables optimized by the PC-MaOEA-PINN. Further modeling and implementation details are discussed in Section 4.2.

In addition, the gas subsystem must satisfy the flow balance constraints of the pipeline network as Eq. (20), and the $G_{i,gas}$ is similarly as $G_{i,ele}$. For modeling purposes, each pipeline segment is treated as an integrated unit. The numerical characteristics of flow and pressure differ between pipelines with GSs and those without.

$$\sum_i^{N_{GAS}} G_{i,gas} \cdot P_i^{gas}(t) = \sum_i^{N_{GAS,LOAD}} P_i^{gas,load}(t) \quad (20)$$

At the nodal level, flow conservation must be enforced, as Eq. (21), the total inflow to a given node must equal the total outflow. This is computed as the sum of inflows from all upstream nodes minus the sum of outflows to all downstream nodes. This constraint ensures continuity of gas transmission and supports the stable operation of the overall network.

$$m_j^g(t) + \sum_{k \in \delta(j)} m_{jk}^g(t) = \sum_{i \in \varphi(j)} m_{ij}^g(t) \quad (21)$$

Additionally, the amount of gas delivered from the gas source to the GT load node must exceed the gas required for GT power generation, therefore, the generation constraint for GTs is Eq. (22). Here, m_i^g is the load of GTs in the gas subsystem.

$$\sum_i^{N_{GT}} m_i^g(t) \leq \sum_i^{N_{GT}} \frac{P_i^{gt}(t)}{\eta_{GT} \cdot Q_{LHV}} \quad (22)$$

3.1.3. The heat and cooling subsystems

They share a similar structural and operational framework, both relying on the transmission of heat or cooling media through pipelines to balance supply and demand. At each node, both the supply and return pipelines are modeled, with the nodal temperature determined by the heat mixing of flows at that location.

The key difference lies in the sources of heat and cooling, that is, EBs on the power load side and CHPs from the gas subsystem. while the cooling subsystem relies on the GTs as CSs. Consequently, GTs must simultaneously support the power and cooling subsystems.

The dynamic behavior of heat/cooling media within the pipelines is described by a temperature propagation equation, formulated as Eq. (23).

$$\frac{\partial T_e}{\partial t} + u \frac{\partial T_e}{\partial x} + \frac{2u}{mhc d_h} (T_e - T_e^a) = \frac{\lambda w}{\rho c} \frac{\partial^2 T_e}{\partial x^2} \quad (23)$$

Similar to the gas network model, the initial temperature is specified, while Dirichlet boundary conditions are imposed via decision variables in the optimization algorithm. Each pipeline is treated as a controllable dispatch unit. The mixed temperature at node j , obtained from the supply and return pipelines, is considered the actual nodal temperature. This temperature must satisfy certain conditions: the inflow temperature must be greater than or equal to the outflow temperature plus the

Table 2

The units of notations.

Notation	Unit	Notation	Unit
P	MWh	Q	MWh
θ	rad	Q_{LHV}	MJ / MWh ³
m_h	kg / h	m_g	Nm ³ / h
pr_{ele}	(/ h	pr_{gas}	(/ Nm ³
pr_{CO_2}	(/ t	ρ_{coal}	t-CO ₂ / MWh
ρ_{gas}	tr-CO ₂ / kg	p	Pa
C_p	MWh / (kg·°C)	c	J / (kg·°C)
T_e	°C	–	–

heat load or losses, and must not exceed a specified upper limit, as represented in Eq. (24).

$$T_e^{load}(t) \leq T_e^{fwd}(t) - T_e^{rev}(t) \leq T_e^{max} \quad (24)$$

For supply pipelines, the nodal temperature is determined by the weighted average of the temperatures at all tail ends connected to the node. For return pipelines, it is calculated based on the temperatures at the head ends of all outbound links, as detailed in Eqs. (25)-(26).

$$T_e^{fwd}(t) = \sum_{i \in \varphi(j)} T_{e_{ij}}(t); T_{e_{jk}}(t) = T_e^{fwd}(t) k \in \delta(j) \quad (25)$$

$$T_e^{rev}(t) = \sum_{i \in \varphi'(j)} T_{e_{ij}}(t); T_{e_{jk}}(t) = T_e^{rev}(t) k \in \delta'(j) \quad (26)$$

At the heat source nodes (HSSs), the heat output must satisfy heat-electrical coupling constraints, as described in Eq. (27).

$$P_i^{CHP,heat}(t) = \alpha_{chp} \cdot P_i^{CHP,ele}(t) \quad (27)$$

$$\sum_i^{N_{CHP}} m_i^g(t) \leq \sum_i^{N_{CHP}} \left(\frac{P_i^{CHP,ele}(t)}{\eta_{g-e}^{CHP}} + \frac{P_i^{CHP,heat}(t)}{\eta_{g-h}^{CHP}} \right) \cdot \frac{1}{Q_{LHV}} \quad (28)$$

$$\sum_i^{N_{EB}} T_e^{fwd}(t) \leq \sum_i^{N_{EB}} \frac{P_i^{eb}(t)}{\eta_{EB}} \cdot c_p \cdot m_h \quad (29)$$

Here, $P_i^{CHP,heat}$ is the heat power output of the CHPs, $P_i^{CHP,ele}$ is electrical output, $m_i^g(t)$ is the gas load of the CHP in the gas subsystem and the α_{chp} is the heat-to-power coupling coefficient. The heat output must also satisfy the equivalent electric demand, as Eqs. (28)-(29).

For the cooling subsystem, the ABS must satisfy the constraint that the absorption chiller is driven by steam generated from the GT. The steam is subsequently converted into cooling energy, as specified in Eq. (30).

$$P_i^{cool}(t) = \eta_{ABS} \cdot P_i^{gt}(t) \quad (30)$$

3.2. Many-objective function setting

This section presents the formulation of objective functions in the MaOCMFM. We define a many-objective optimization problem involving five conflicting objectives: economic cost, system profit, RES utilization rate, operational stability, and user comfort. Table 2 summarizes the key variables and their respective units.

3.2.1. Economic cost

The economic cost objective captures the operational cost contributions from all four subsystems: electricity, gas, heat, and cooling networks. Let F^{cost} denote the unit cost. For the power subsystem, the generation cost of conventional THs is modeled as Eq. (31).

$$F_{fuel}^{cost}(t) = \sum_i^{N_F} (\alpha_i^f \cdot P_i^{fuel}(t)^2 + \beta_i^f \cdot P_i^{fuel}(t) + \gamma_i) \quad (31)$$

Where $\alpha_i^f, \beta_i^f, \gamma_i$ are cost coefficients and P_i^{fuel} is the power output. The ramping costs for THs are Eq. (32).

$$F_{ramp}^{cost}(t) = \sum_{i=1}^{N_F} \left(\beta_i^{up} \cdot \max(0, \frac{dP_i^{fuel}(t)}{dt}) + \beta_i^{down} \cdot \min(0, \frac{dP_i^{fuel}(t)}{dt}) \right) \quad (32)$$

Eq. (33) defines the power loss cost of the electrical network, which is calculated based on the active power loss across each transmission line.

$$F_{ele}^{loss} = pr_{ele}(t) \cdot \sum_i^{N_{EBR}} (P_i^f(t) + P_i^l(t)) \quad (33)$$

Here, P^f and P^l are the power of from nodes and to nodes, respectively. pr_{ele} denoted time-of-use electricity price. Specifically, the loss on each branch is obtained as the algebraic sum of power injections at both ends. The total power loss cost is then determined by aggregating the losses across all branches in the system.

The energy loss during BESSs charging/discharging is expressed as Eq. (34).

$$F_{bess}^{loss}(t) = pr_{ele}(t) \cdot \left(\sum_{i=1}^{N_{BESS}} P_i^{bess, ch}(t) \cdot (1 - \eta_{ch}) + \frac{\sum_{i=1}^{N_{BESS}} P_i^{bess, dis}(t) \cdot (1 - \eta_{dis})}{\eta_{dis}} \right) \quad (34)$$

For grid-connected IES, the power purchase cost is Eq. (35).

$$F_{buy}^{cost}(t) = pr_{ele}(t) \cdot \sum_i^{N_{BUY}} P_i^{buy}(t) \quad (35)$$

In the gas subsystem, the IES operation cost consists of the gas purchase cost from gas wells. Given that the gas flow dynamics are modeled using one-dimensional compressible fluid dynamics equations, the mass flow rate is adopted as the primary flow variable.

The unit gas purchase cost is calculated as shown in Eq. (36), where pr_{gas} represent the price of natural gas.

$$F_{gas}^{cost}(t) = pr_{gas} \cdot \sum_i^{N_{GS}} m_i^g(t) \quad (36)$$

The operational cost of the heat subsystem is determined by the heat output of the HSs, including CHPs and EBs. Since both types of HSs simultaneously serve the power subsystem and comply with the equivalent electric load constraints, they are therefore not calculated separately within the thermal network.

In the cooling subsystem, the CSs is ABSs, which is powered by the exhaust gas from the GTs. Within IES, the exhaust gas from the GTs can be utilized not only to drive the ABS but also to supply the heat subsystem via a heat recovery boiler (HRB), enabling multi-energy synergy.

Therefore, the opportunity cost of the CSs is considered in the modeling to reflect the competing utilization of GTs exhaust between cooling and heating, as formulated in Eq. (37).

$$F_{cool}^{cost}(t) = \left(\frac{pr_{gas}}{\eta_{GT} \cdot \eta_{HRSG}} - \frac{1}{2} \left(\frac{pr_{gas}}{\eta_{CHP}} + \frac{pr_{ele}(t)}{\eta_{EB}} \right) \right) \cdot \sum_{i=1}^{N_{ABS}} \frac{P_i^{cool}(t)}{\eta_{ABS}} \quad (37)$$

Here, $pr_{gas}/(\eta_{GT} \cdot \eta_{HRSG})$ denotes the equivalent price of exhaust gas, calculated proportionally based on its utilization efficiency relative to natural gas, while $(pr_{gas}/\eta_{CHP} + pr_{ele}(t)/\eta_{EB})/2$ represents the unit price of the equivalent thermal power.

Based on the cost components of the aforementioned subsystems, the overall cost objective function is formulated as Eq. (38).

$$F^{cost}(t) = F_{fuel}^{cost}(t) + F_{ramp}^{cost}(t) + F_{ele}^{loss}(t) + F_{bess}^{loss}(t) + F_{buy}^{cost}(t) + F_{gas}^{cost}(t) + F_{cool}^{cost}(t) \quad (38)$$

3.2.2. System profit

The system profit takes into account the carbon emission revenue obtained by the IES through participation in the carbon market, as shown in Eq. (39).

$$F^{profit}(t) = \left(C_{quto}(t) - \sum_{i=1}^{N_F} \eta_{fuel} \cdot P_i^{fuel}(t) \cdot \rho_{coal} + \sum_{i=1}^{N_g} \eta_{gas} \cdot P_i^{gt}(t) \cdot \rho_{gas} + \sum_{i=1}^{N_{CHP}} \eta_{CHP} \cdot P_i^{CHP}(t) \cdot \rho_{gas} \right) \cdot pr_{CO_2}(t) \quad (39)$$

Here, C_{quto} denotes the carbon quota revenue earned by the IES through participation in the carbon market, which is determined by the difference in CO_2 emissions between the IES operation and the combined emissions from conventional THs, GTs, and CHPs, pr_{CO_2} is the CO_2 price in the carbon market, and ρ_{coal} and ρ_{gas} are the carbon emission factors of coal and gas, respectively.

3.2.3. RES utilization rate

Moreover, since wind and photovoltaic power are typical non-dispatchable resources, the utilization rate of renewable energy sources (RES) is evaluated by the ratio between the actual outputs, $P^{PV, actual}$ and $P^{WIN, actual}$, and the forecasted outputs, $P^{PV, predict}$ and $P^{WIN, predict}$, of photovoltaic units (PVs) and wind turbines (WINDs) within the integrated energy system (IES), as defined in Eq. (40).

$$F^{res}(t) = \frac{\sum_i^{N_{PV+WI}} P_i^{PV, actual}(t) + P_i^{WIN, actual}(t)}{\sum_i^{N_{PV+WI}} P_i^{PV, predict}(t) + P_i^{WIN, predict}(t)} \quad (40)$$

3.2.4. Operational stability

Operational stability includes minimizing voltage deviation across power grid nodes, ensuring smooth power output from the BESSs, suppressing heat overshoot from HSs, minimizing gas flow variation from GSs, and reducing power fluctuations of CSs.

$$F^{sta}(t) = \sum_i^{N_T} \frac{dP_i(t)}{dt} \quad (41)$$

The first-order gradients (slopes) of each node are introduced as a stability indicator, as shown in Eq. (41). A lower gradient indicates smoother variations and enhanced operational stability. Therefore, the stability objective of the IES is formulated as minimizing the gradients of power or flow profiles across all critical nodes. The primary purpose of this stability metric is to ensure the security margin of the power grid, which is conceptually similar to the calculation principles of voltage stability margin, frequency stability, and transient stability—each of which is essentially derived from differences between two operating states. In our case, since the decision variables are represented by piecewise continuous spline functions, gradients naturally serve as an appropriate measure of stability.

3.2.5. User comfort

User comfort is quantified in this study using the Predicted Mean Vote (PMV) index, as defined in Eqs. (42)–(43).

$$F^{comfort}(t) = \begin{cases} |\lambda_i^{PMV}(t)| - 0.5, & \text{if } |\lambda_i^{PMV}(t)| \geq 0.5, \\ 0, & \text{if } |\lambda_i^{PMV}(t)| < 0.5 \end{cases} \quad (42)$$

$$\lambda_i^{PMV}(t) = f(T_{e,i}^I(t), RH(t), v_{air}(t), met, clo) \quad (43)$$

User comfort objective is defined as the absolute deviation of the PMV value from 0.5, where a smaller value of the objective function $F^{comfort}$ indicates a higher level of perceived thermal comfort. Here, λ_{PMV} denotes the PMV value, which is a nonlinear function of indoor temperature ($T_{e,i}^I$), relative humidity (RH), air velocity (v_{air}), metabolic rate (met), and clothing insulation (clo). The detailed calculation method follows the approach described in reference Cheung et al. (2019). The $T_{e,i}^I$ is derived from the corresponding node i in the heat subsystem.

3.3. Model setting and encoding strategies

By jointly considering Eqs. (38)–(42) and the associated constraints, the MaOCMFM problem is formulated as a many-objective optimization problem, as shown in Eq. (44).

$$F(t) = \{ F^{cost}(t), F^{profit}(t), F^{res}(t), F^{sta}(t), F^{comfort}(t) \}, \quad t \in [0, T], \text{ s.t. (1) } \sim (30) \quad (44)$$

In MaOCMFM, all optimization objectives are modeled in the form of integrals over the scheduling horizon $[0, T]$, in order to comprehensively

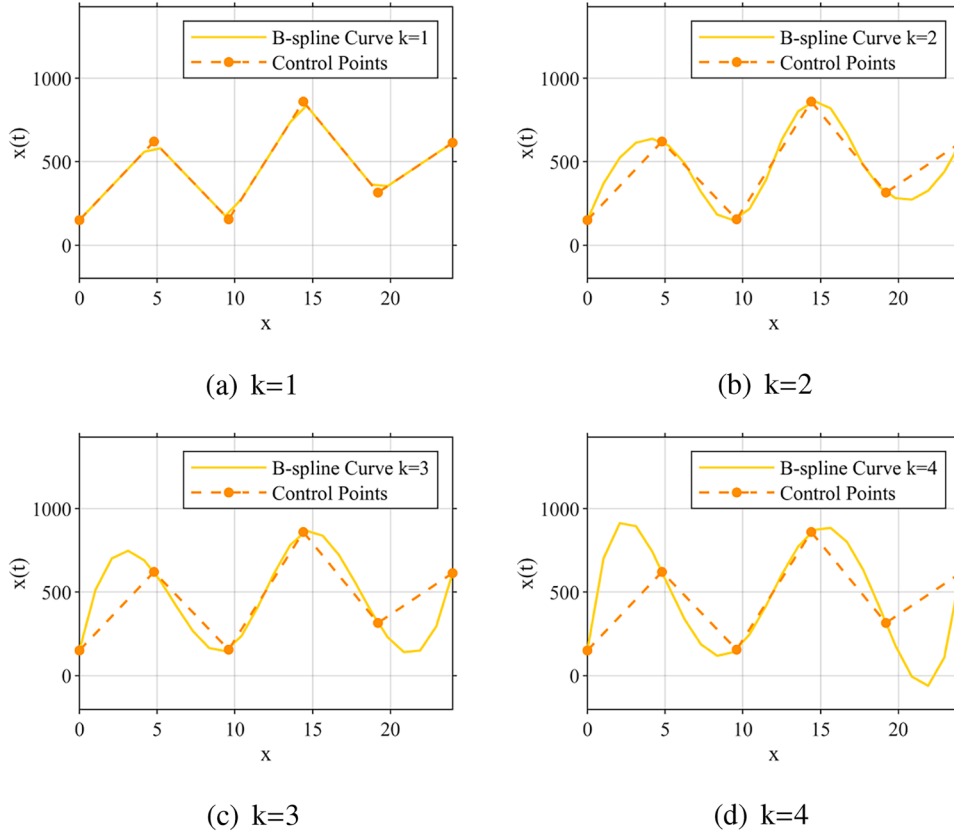


Fig. 2. Schematic diagram of the B-spline function under same control points, $k = 1, 2, 3, 4$.

capture the dynamic performance of the integrated energy system (Cui et al., 2023).

To solve this problem, the individuals in the PC-MaOEA-PINN are encoded using cubic B-spline functions, which provide a smooth representation of the temporal evolution of scheduling variables. The general form of a cubic B-spline function is expressed as Eq. (45):

$$s(t) = \sum_{i=1}^n \alpha_i B_i^3(t) \quad (45)$$

where $s(t)$ denotes the decision variable at time t , represents the control point coefficient, is the i th cubic B-spline basis function, and n is the number of control nodes. The knot vector is uniformly sampled over the interval $[0, T]$.

The recursive definition of the cubic B-spline basis functions $B_i^3(t)$ is given in Eqs. (46)-(47):

$$B_i^0 = \begin{cases} 1, & t_i \leq t < t_{i+1}, \\ 0, & \text{otherwise} \end{cases} \quad (46)$$

$$B_i^k(t) = \frac{t - t_i}{t_{i+k} - t_i} B_i^{k-1}(t) + \frac{t_{i+k+1} - t}{t_{i+k+1} - t_{i+1}} B_{i+1}^{k-1}(t) \quad (47)$$

Where $\{t_i\}_{i=1}^{n+k+1}$ is a non-decreasing sequence of real numbers defined over the interval $[0, T]$, and k denotes the degree of the B-spline basis function.

Fig. 2 illustrates a comparison of B-spline fitting curves under different orders. When $k = 3$, the B-spline fitting curve exhibits better smoothness, which enables it to maintain good approximation accuracy while effectively suppressing overfitting. This makes it well-suited for continuous modeling and interpolation of system scheduling variables.

Fig. 3 illustrates the fitting behavior of cubic B-spline functions under different control point configurations. Each set of control points corresponds to a specific power output scenario of THs.

Using cubic B-spline functions, the encoding scheme for each individual is summarized in Fig. 4. The encoded decision variables, from left

to right in Fig. 4, are the active and reactive power of generation nodes, the voltage magnitudes of load nodes, the phase angles of all nodes in the electrical subsystem, the gas pressures of all gas subsystem nodes, the flow rates at GSs, the supply water temperatures of HSs, the return water temperatures of all nodes in the heat subsystem and cooling subsystem, and the power of BESS.

Each individual contains all decision variables required for scheduling. For each gene, control points are uniformly sampled over the scheduling interval $[0, T]$ and serve as the decision variables. This encoding scheme ensures that sampling can be performed at the boundaries of the PDEs, facilitating the subsequent training of the PINNSM.

4. Probabilistic contributing many-objective evolutionary algorithm enhanced by physics-informed neural network surrogate model

This subsection provides a detailed description of the algorithmic implementation of PINNSM and PC-MaOEA-PINN. In this work, PINNSM are separately trained for the heat subsystem, the cooling subsystem, and the gas subsystem. During the evolutionary phase, these models will be integrated into the fitness value of objective function and constraint violations. Fig. 5 illustrates the PC-MaOEA-PINN process.

The specific architecture of PINNSM is detailed in Section 4.1, while the trained model is utilized during the iterative process of the algorithm to evaluate the fitness and Constraint value. The corresponding algorithmic details are provided in Section 4.2.

4.1. PINNSM

The proposed PINNSM employs a surrogate-assisted strategy by embedding PINNs into the evolutionary algorithm for evaluating PDE-related fitness values and constraint violations. Unlike conventional surrogate models in evolutionary algorithms, PINNSM does not directly

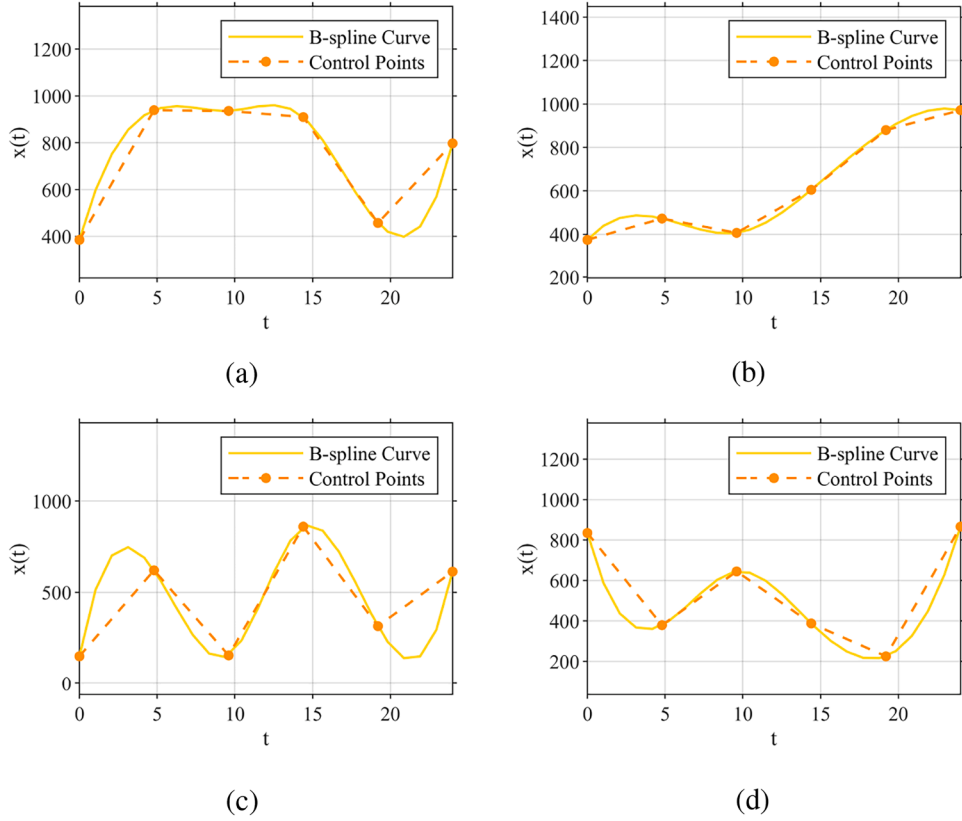


Fig. 3. Schematic diagram of the B-spline function under different control points.

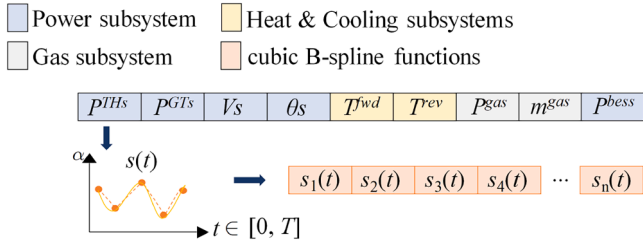


Fig. 4. The encoding scheme for each individual.

approximate the objectives or constraints. Instead, it solves the underlying PDE, and the fitness and constraint violations are subsequently computed from the solution $u(x, t)$. Thus, the computational burden arises not from the evaluation itself, but from obtaining accurate PDE solutions (Wen et al., 2024).

To mitigate this cost, PINNSM is pre-trained using datasets constructed by randomly sampling boundary conditions in the decision space. For each sampled condition, PDE solutions are computed on fine-grained grids via numerical schemes, which serve as labels. The inputs of PINNSM consist of boundary conditions and spatiotemporal points, while the outputs approximate the PDE solutions. Since the sampled data do not cover all possible conditions, PINNSM leverages physics-informed learning to generalize, requiring only a small number of labeled points to achieve high accuracy.

As illustrated in Fig. 6, the traditional grid-based method (Fig. 6(a)) demands dense discretization, resulting in prohibitive computational cost, whereas PINNs (Fig. 6(b)) require only sparse sampling. This efficiency arises because PINNs embed the PDE form, boundary, and initial conditions into the training, where most residual points are enforced through gradient-based approximations rather than explicit numerical computation. Consequently, PINNSM significantly reduces the number

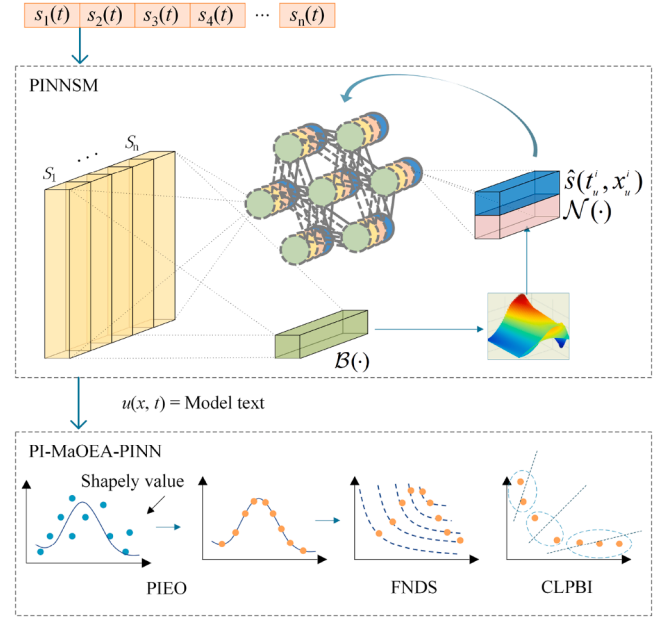


Fig. 5. Schematic diagram of PC-MaOEA-PINN.

of expensive grid-based calculations while maintaining high-fidelity approximations.

The proposed PINNSM is an extension of PINNs. Firstly, PINNSM is specifically designed as a model embedded into evolutionary algorithms to compute the corresponding PDE solutions. From this perspective, PINNSM represents an application of PINNs in the field of evolutionary computation. Secondly, in standard PINNs, parts of the PDE

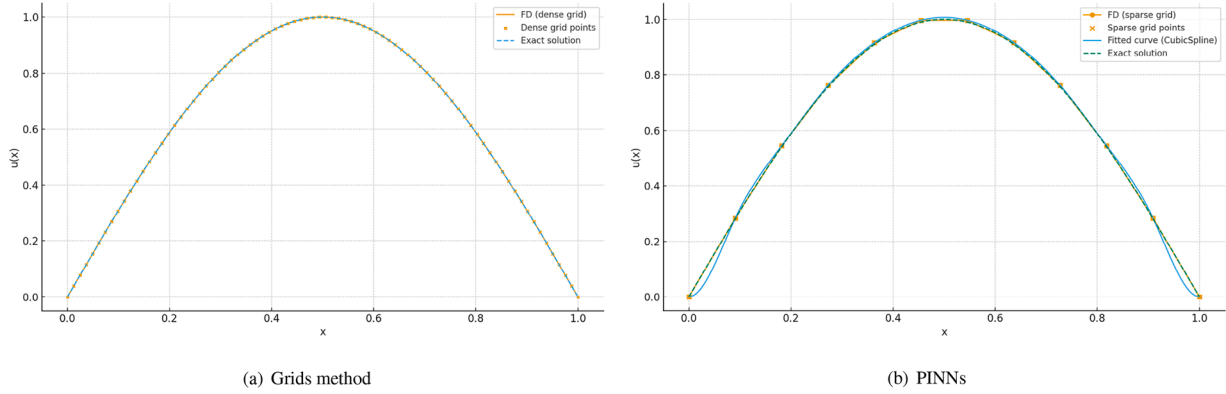


Fig. 6. Comparison of sampling points between the grid method and PINNs.

numerical solutions, initial conditions, and boundary conditions are explicitly known. In contrast, in PINNSM, the boundary conditions are not directly available. This creates a fundamental challenge: the model must learn to approximate PDE solutions without relying on pre-specified boundary information. In this work, we address this by constructing training data through a concatenation strategy, where boundary conditions and partial numerical solutions under those conditions are combined to form the dataset (Lin et al., 2025). The backbone network remains consistent with the original PINNs architecture. While the resulting PDE solutions may not achieve the same accuracy as PINNs with explicitly defined boundary conditions, this approach avoids the need for computationally expensive grid-based PDE solvers. Moreover, the obtained solutions are sufficiently useful for evolutionary optimization, representing a practical trade-off.

$$\begin{aligned} \mathcal{N}(t, x; u(t, x)) &= 0, \quad t \in [0, T], \quad x \in D, \\ I(x; u(0, x)) &= 0, \quad x \in D, \\ B(t, x; u(t, x)) &= 0, \quad t \in [0, T], \quad x \in \partial D. \end{aligned} \quad (48)$$

Mathematically, a PDE can be expressed as follows in Eq. (48) (Zha et al., 2022), where $\mathcal{N}(\cdot)$ denotes the differential operator that encapsulates the space-time differential terms, $u(x, t)$ represents the solution to the PDE, $I(\cdot)$ represents the initial condition, and $B(\cdot)$ denotes the boundary condition.

In the PINNSM, $\mathcal{N}(\cdot)$, $I(\cdot)$, and $B(\cdot)$ are integrated into a new loss function designed to capture relevant physical features from the data, as shown in Eq. (49).

$$\begin{aligned} loss(S(w, b)) &= loss_{data}(S(w, b)) + loss_{psy}(S(w, b)) \\ &= \|u(t_f^i, x_u^i) - \hat{u}(t_f^i, x_u^i)\| + \|\mathcal{N}(t_f^i, x_f^i)\| + \|B(t_b^i, x_b^i)\| \\ &\quad + \|I(0, x_0^i)\| \end{aligned} \quad (49)$$

Where the $loss_{data}$ is computed using MSE in the PINNSM, and $\|\mathcal{N}(t_f^i, x_f^i)\|$, $\|B(t_b^i, x_b^i)\|$, and $\|I(0, x_0^i)\|$ are the respective coefficients for the PDE loss, boundary condition loss, and initial condition loss, $S(w, b)$ is the weights and biases in neural networks. Notably, the $loss_{psy}$ does not require data labels, only sampling points in both the time and spatial domains. $\{t_u^i, x_u^i\}_{i=1}^{N_u}$ denotes labelled data points, or the original data, with a total of N_u , and $\{t_f^i, x_f^i\}_{i=1}^{N_f}$ represents the sampled data points in the time and spatial domains, totaling N_f .

In the PC-MaOEA-PINN, $\|B(t_b^i, x_b^i)\|$ are sampled from the B-spline function $s(t)$. Therefore, for the gas subsystem and heat & cooling subsystem corresponding to the PDEs, the expressions are as follows in

Eq. (50)–(51).

$$\begin{cases} \frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial p}{\partial x} + g(\rho - \rho_a) \sin \alpha + \frac{\lambda}{d} \frac{\rho u^2}{2} = 0, \\ \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0, \\ u(x, t) = \sum \alpha_i^u B_i^3(t), \\ p(x, t) = \sum \alpha_i^p B_i^3(t), \\ u(x, 0) = m^g(x, 0), \quad p(x, 0) = P^{gas}(x, 0). \end{cases} \quad (50)$$

$$\begin{cases} \frac{\partial T e}{\partial t} + v \frac{\partial T e}{\partial x} + \frac{2u}{m^h c d_h} (T e - T e^a) = \frac{\lambda^w}{\rho c} \frac{\partial^2 T e}{\partial x^2}, \\ T e(x, t) = \sum \alpha_i^{Te} B_i^3(t), \\ T e(x, 0) = T e_i^I. \end{cases} \quad (51)$$

Here, $B(\cdot)$ is captured from each generation of evolved individuals, while $I(\cdot)$ is provided based on empirical data. The loss function of PINNSM is expressed as follows in Eq. (52):

$$\begin{aligned} F_{loss} &= \|s(t_u^i, x_u^i) - \hat{s}(t_u^i, x_u^i)\| + \|\mathcal{N}(s(t_f^i, x_f^i))\| \\ &\quad + \|B(s(t_b^i, x_b^i))\| + \|I(s(t_0^i, x_0^i))\| \end{aligned} \quad (52)$$

The PINNSM architecture is designed as a fully connected neural network tailored for regression tasks. The input consists of encoded features of individuals, and the output is a vector aligned with the number of individuals. Physical information of the problem is incorporated into the loss function to guide the optimization of network weights. Essentially, this approach leverages the PINN framework to model the relationship among state data, boundary conditions, and the fitness values of objective function (Cai et al., 2021; Cui et al., 2018). Accordingly, the fitness value can be interpreted as a weighted combination of the objective function and the solution to the PDE. The pre-training process of PINNSM is shown in Algorithm 1.

In this paper, each surrogate model is constructed independently with respect to a specific PDE (Xu et al., 2025), ensuring that the physical constraints and solution characteristics of the corresponding PDE are explicitly embedded into the training process. This independent construction not only improves the fidelity of the surrogate models to their target PDEs, but also facilitates the subsequent integration of multiple surrogates in complex optimization scenarios.

4.2. PC-MaOEA-PINN

The PC-MaOEA-PINN is Regarding the design of the dual-population mechanism, its primary purpose is to better address constraints. The two populations consist of a constrained population and an unconstrained population. The unconstrained population typically exhibits stronger

Algorithm 1 PINNSM training procedure.

Input: F (PDE expression), I (Initial condition), N_s (Number of training samples), η (Learning rate), max_epoch (Maximum training iterations)
Output: H (Trained PINNSM)

- 1: Initialize the PINN model H ;
- 2: Randomly sample N individuals as boundary conditions;
- 3: For each boundary condition $s(t)$, sample N_b points, generating a total of $N_b \times N$ boundary data points;
- 4: Compute the numerical solutions $u(x, t)$ of the PDE using finite difference schemes on a fine grid;
- 5: Construct the training dataset $u(x_d, t_d) \cup s(t_b)$;
- 6: **for** $epoch = 1$ to max_epoch **do**
- 7: Sample N_i points at the initial time $\{x_i, 0\}$;
- 8: Randomly sample N_f collocation points within the computational domain $\{x_f, t_f\}$;
- 9: Compute the physics loss and data loss (see Eq. (52));
- 10: Update the network parameters of H using gradient-based optimization with learning rate η ;
- 11: **end for**

convergence, as it is not restricted by feasibility. Consequently, well-converged individuals in this population may lie either inside or outside the feasible region. During population migration, selectively transferring well-converged individuals into the constrained population provides new evolutionary directions for the latter, thereby steering the overall algorithm toward feasible regions with improved convergence.

The design of the dual-population mechanism in PC-MaOEA-PINN primarily aims to better address constraints. Specifically, two populations are maintained: a constrained population and an unconstrained population. The unconstrained population generally exhibits stronger convergence, as it is not limited by feasibility requirements. As a result, well-converged individuals in this population may lie either inside or outside the feasible region. During population migration, selectively transferring well-converged individuals into the constrained population introduces new evolutionary directions, thereby guiding the overall algorithm toward feasible regions with improved convergence. The overall framework of PC-MaOEA-PINN is summarized in Algorithm 2.

In the proposed PC-MaOEA-PINN, two subpopulations, namely the constrained population S_1 and the unconstrained population S_2 , are simultaneously generated during the optimization process, together with the initialization of reference vectors. The algorithm can be divided into three main phases. In the first phase, when $FES < \text{firstFES}$, both S_1 and S_2 evolve under the combined use of the Genetic Algorithm (GA) operator and the Probabilistically Contributing Evolutionary Operator (PCEO) (see Steps 8–9), which are employed to generate offspring. The PCEO is constructed on a multivariate Gaussian distribution model and incorporates a cooperative game mechanism by computing the Shapley values of each decision variable (Fang et al., 2025). According to their contributions, probabilistic weights are assigned to different positions, and offspring individuals are sampled from the resulting weighted distribution, which enables the algorithm to concentrate its search on the most critical evolutionary regions.

In the subsequent parent-offspring merging stage, both CS_1 and CS_2 are set to a population size of $3N$. A pre-trained PINNSM model H is then employed to evaluate each individual with respect to both constraint violations and objective values. It should be emphasized that PINNSM does not directly approximate the constraint violations or the objective values; instead, its primary function is to solve the PDE solution $u(x, t)$ associated with the constraints. Once the individual solution $u(x, t)$ is obtained, the constraint violation can be computed in a conventional manner based on $u(x, t)$. After obtaining the objective and constraint violation values of all individuals, a novel environmental selection operator (Step 15) is introduced. This operator integrates

Algorithm 2 PC-MaOEA-PINN.

Input: N (Population size), firstFES (Number of first-stage function evaluations), maxFES (Maximum function evaluations), H (PINNSM), M (Number of objectives) **Output:** P_1 (Optimal solution set)

- 1: Initialize population S_1 as constrained population, S_2 as unconstrained population;
- 2: $W \leftarrow \text{ReferencePointGeneration}(N, M)$
- 3: $FES \leftarrow 2N$;
- 4: **while** $FES < \text{maxFES}$ **do**
- 5: **if** $FES < \text{firstFES}$ **then**
- 6: $S_1 \leftarrow \text{Mating Selection}(S_1)$, $S_2 \leftarrow \text{Mating Selection}(S_2)$;
- 7: **for** $i = 1$ to 2 **do**
- 8: $O_1 = \text{Generate } N/2 \text{ offspring from } S_i \text{ using standard GA operator}$;
- 9: $O_2 = \text{Generate } N/2 \text{ offspring from } S_i \text{ using PCEO operator}$; (see Algorithm 2)
- 10: **end for**
- 11: $CS_1 = S_1 \cup O_1^{GA} \cup O_1^{PCEO} \cup O_2^{GA} \cup O_2^{PCEO}$, $CS_2 = S_2 \cup O_1^{GA} \cup O_1^{PCEO} \cup O_2^{GA} \cup O_2^{PCEO}$;
- 12: **for each** $s(x, t) \in CS_1, CS_2$ **do**
- 13: Calculate the constraint violation and objective values using H ;
- 14: **end for**
- 15: $S_1, S_2 \leftarrow \text{Environment Selection}(CS_1, CS_2, W)$; (see Algorithm 3)
- 16: $FES \leftarrow FES + 2N$;
- 17: **else if** $FES = \text{firstFES}$ **then**
- 18: Type 1–6 $\leftarrow$ Define the knowledge transfer strategies;
- 19: **else if** $FES > \text{firstFES}$ **then**
- 20: Repeat steps (7)–(8) to generate O_1 ;
- 21: **if** Type 1 or Type 2 is true **then**
- 22: Select the GA and Transfer operator with equal probability to generate O_2 ;
- 23: **else if** Type 3 or Type 4 or Type 5 is true **then**
- 24: Execute the PCEO operator alone to generate O_2 ;
- 25: **else if** Type 6 is true **then**
- 26: Select the GA, Transfer and PCEO operator with equal probability to generate O_2 ;
- 27: **end if**
- 28: $CS_1 = S_1 \cup O_1 \cup O_2$, $CS_2 = S_2 \cup O_1 \cup O_2$;
- 29: $S_1, S_2 \leftarrow \text{Environment Selection}(CS_1, CS_2, W)$;
- 30: $FES \leftarrow FES + 2N$;
- 31: **end if**
- 32: **end while**

a function-based non-dominated sorting method (CFNDS) with a clustering and PBI-distance-based strategy (CLPBI), thereby enabling the selection of elite individuals from different clusters and effectively mitigating the loss of selection pressure in many-objective optimization scenarios.

In the second phase, when $FES = \text{firstFES}$, the algorithm classifies the current two populations according to a predefined knowledge transfer strategy. Specifically, the knowledge transfer mechanism adopted here follows that in the URCMO algorithm (Liang et al., 2022), as illustrated in Fig. 7.

In Fig. 7, the red dashed line represents the Pareto front of the unconstrained subpopulation S_2 , while the blue solid line corresponds to the Pareto front of the constrained subpopulation S_1 . The URCMO framework categorizes their relationship into six cases: Types1 and2 indicate complete overlap between the Pareto fronts of S_1 and S_2 ; Types3–5 correspond to complete separation; and Type6 reflects partial overlap. The classification criterion is based on the proportion of non-dominated solutions from each subpopulation within the overall set of non-dominated solutions in $S_1 \cup S_2$, as formulated in Eq. (53). According to these

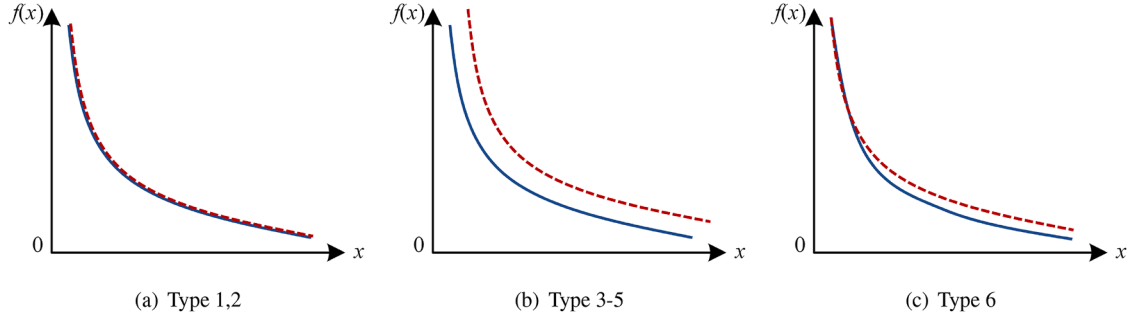


Fig. 7. Six types of Pareto frontiers exist in both the constrained population and the unconstrained population.

proportions, URCMO distinguishes the six types of relationships.

$$\rho_i = \frac{|ND(S_i) \cap ND(S_1 \cup S_2)|}{|ND(S_1 \cup S_2)|}, \quad i \in \{1, 2\}, \quad (53)$$

Where $ND(\cdot)$ denotes the set of non-dominated solutions. In the third phase, when $FES > \text{firstFEs}$, Steps 7–8 are first repeated to generate an intermediate offspring set O_1 . Then, different combinations of evolutionary operators are applied depending on the classification type. For Type 1 and Type 2, since the Pareto fronts of S_1 and S_2 are completely coincident, the knowledge contained in the unconstrained population S_2 can be effectively exploited by S_1 . In this case, the GA operator and a Transfer operator are employed, where the Transfer operator is defined as a stochastic migration operator, as Eq. (54).

$$x_i^{t+1} = \begin{cases} x_j^t, & \text{if } r_1 = r_2, \\ x_i^t, & \text{if } r_1 \neq r_2, \end{cases} \quad (54)$$

Where r_1 and r_2 are two independently generated random numbers, x_i^t denotes the position of individual i in generation t , and x_j^t represents an individual selected from the other population. This mechanism implies that the state of x_i will be migrated to that of x_j only when the two random numbers are equal; otherwise, x_i remains unchanged. Such a design introduces stochasticity into the transfer process, ensuring that knowledge exchange between subpopulations occurs in a probabilistic manner while maintaining population diversity.

For Type 3, Type 4, and Type 5, the Pareto fronts of S_1 and S_2 are regarded as completely separated, and thus S_2 evolves independently using only the PCEO operator, without importing individuals from S_1 . This design ensures that S_2 focuses more on distributional shifts that contribute to the evolutionary process. If, during subsequent generations, the Pareto fronts of the two populations become partially or fully coincident, only individuals that significantly enhance convergence will be transferred. For Type 6, where partial overlap between the Pareto fronts exists, the algorithm adopts GA, PCEO, and Transfer operators with equal probability, thereby achieving a balance between convergence and feasibility in the evolutionary process.

4.2.1. PCEO

The PCEO operator is employed to update the distribution of the probabilistic model, as described in Algorithm 3. By adaptively adjusting the search distribution according to importance measures, this strategy strengthens the algorithm's ability to escape local optima and improves the overall exploration capability. The PCEO operator employs Shapley values to evaluate the contribution of each gene position (i.e., the control points of the B-spline function) to the overall scheduling strategy. The motivation is to precisely identify the impact of each gene locus, thereby enabling targeted adjustments rather than relying purely on randomness. Specifically, we group subsystems, compute the gradient of each gene position with respect to time, and measure whether a gene position plays a critical role in influencing the objective function by examining the rate of gradient variation. This is made possible by the piecewise continuity property of B-spline functions, which allows gradient computation.

Algorithm 3 PCEO operator.

Input: S (parent population), N (population size), D (dimension of decision variables), K (number of groups) **Output:** Q (offspring population)

```

1: for  $i = 1$  to  $N$  do
2:   for  $j = 1$  to  $D$  do
3:     for  $k = 1$  to  $K$  do
4:        $\rho_k \leftarrow$  Shapley value of the  $k$ th group, as in Equation (57);
5:     end for
6:      $\mu_j \leftarrow$  Mean value of the  $j$ th gene, as in Equation (55);
7:   end for
8:    $id\_x = \text{sort}(\Theta, \text{ascend})$ ,  $\Theta = \rho_1, \dots, \rho_K$ ;
9:    $\sigma_k \leftarrow$  Variance of the  $k$ th group, as in Equation (56);
10:   $\sigma_k$  Update  $\sigma_k$  as in Equation (57);
11:   $s_{ij} \sim \mathcal{N}(\mu_j, \sigma_j)$ ;
12: end for

```

Here, s_{ij} denotes the value of the j th gene in the i th individual. μ_j represents the mean value of all individuals at position j , while σ_j is a matrix representing the covariance among the individuals in the population S at position j , as defined in Eqs. (55)–(56).

$$\mu_j = \frac{1}{N} \sum_{i=1}^N s_{ij} \quad (55)$$

$$\sigma = \frac{1}{N} \sum_{i=1}^N (s_i - \mu_j) \cdot (s_i - \mu_j)^T \quad (56)$$

We employed a multivariate Gaussian distribution to characterize the sampling criterion. The mean value is updated using the current mean value of the population S . The set S contains the elite individuals selected by CLPBI, which is analogous to the crossover operation in genetic algorithms (GA). This mechanism ensures that PCEO performs sampling around the updated mean of the new distribution.

Next, we partition the decision variables into K groups. To capture the continuity and correlation among scheduling time steps, the model assigns a B-spline function defined over t time points to each group. Subsequently, we calculate the associated covariance matrix σ_k , and then update σ_k by computing the weighted sum of the normalized marginal contributions ρ_k of each group.

$$\rho_k = \sum_{S' \subseteq S \setminus s_{kj}} \frac{|S'|! (|S|! - |S'|)!}{|S|!} [\text{IMP}(S' \cup \{s_{kj}\}) - \text{IMP}(S')] \quad (57)$$

Eq. (57) presents the calculation formula for the Shapley value ρ_k of group k . $S \setminus \{s_{kj}\}$ denotes the group k excluding the j th gene, and thus $[\text{IMP}(S' \cup \{s_{kj}\}) - \text{IMP}(S')]$ represents the marginal contribution of gene s_{kj} in group k . IMP represents the importance of the j th gene s_{kj} to group k .

Since $s(t)$ is a B-spline function that is differentiable within its domain, the chain rule applies, and it follows that the marginal contribution can be expressed as $\nabla_s f \cdot \nabla_t s$ where $\nabla_t s$ denotes the derivative of s with respect to t . A larger variation in $\nabla_t s$ implies a greater change in the objective function $f(s(t), t)$. Therefore, the importance IMP is defined as

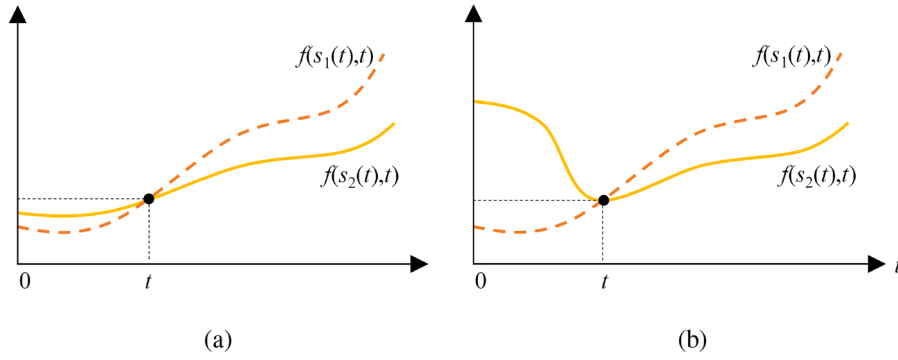


Fig. 8. Two typical scenarios in the objective evaluation.

shown in Eq. (58).

$$IMP = \frac{1}{N} \sum_i^N \nabla_i s_{ik}(t) \quad (58)$$

where s_{ik} represents the k th group of the i th individual, and the importance measure IMP is quantified by calculating the sum of the gradients of $\nabla_i s$. This implies that when the probabilistic model samples at positions with relatively small values of ρ_k , the elite set S from the current population may provide insufficient exploratory guidance in the objective space, thus requiring an increase in sampling probability.

$$\sigma_k = \sigma_k^S + \gamma \cdot \Theta \cdot (\sigma_k^S - \sigma_k) \cdot \gamma \cdot \Theta^T \quad (59)$$

To address this, in Step 10 of Algorithm 3, the ρ_k of each group are sorted in ascending order. The indices corresponding to the smallest ρ_k are selected to construct an index matrix Θ . The matrix Θ is a 0–1 diagonal matrix, and the covariance update is then performed according to Eq. (59), where the parameter $(\sigma_k^S - \sigma_k)$ controls the shift of the covariance towards the current generation. Specifically, larger updates are applied in regions with smaller ρ_k , allowing the population to adapt more aggressively in those areas. This procedure is analogous to a mutation operation, introducing greater exploration freedom into the search process.

4.2.2. Environment selection

Environmental selection incorporates two new strategies. The first is the CLPBI strategy, which integrates the Penalty Boundary Intersection (PBI) metric with a clustering-based individual selection approach. Within this strategy, we embed the proposed CFNDS ranking criterion to account for the characteristics associated with scheduling time. The overall environmental selection process is outlined in Algorithm 4.

In Step 1 of Algorithm 4, the corresponding reference vectors are generated from the given reference points. For each population, the environmental selection strategy is applied iteratively. Specifically, the merged population CS is first sorted according to the non-dominated criterion defined by CFNDS. The definition of CFNDS is given as follows.

Regarding the minimization problem, as Eq. (60).

$$\min f_M(s(x, t), t) = \min \{f_1(s(x, t), t), \dots, f_m(s(x, t), t)\} \quad (60)$$

$$s \in R^n, x = \{x_1, x_2, \dots, x_n\}, t \in [0, T]$$

Here, $s(t)$ is a time B-spline function, and x is the control point corresponding to s . When decision variables are represented using B-spline functions in the algorithm, we must account for a particular situation during objective function comparison, as illustrated in Fig. 8.

In Fig. 8(a) and (b), within the time interval $[0, t]$, the objective $f(s_2(t), t)$ is superior to $f(s_1(t), t)$; whereas in the interval $[t, \text{end}]$, $f(s_2(t), t)$ outperforms $f(s_1(t), t)$. In the execution of evolutionary algorithms, both scenarios are treated as non-dominated solutions. This leads to a large number of non-dominated solutions in many-objective optimization problems, making it difficult to generate high-quality populations.

Algorithm 4 Environmental selection.

Input: W (weight/reference vectors), CS_1 (Combined population 1), CS_2 (Combined population 2) **Output:** S_1, S_2

```

1: for  $i = 1$  to 2 do
2:    $F = [F_1, F_2, \dots, F_L] \leftarrow$  Apply the CFNDS ranking to  $CS_i$ ;
3:   if  $|F_{L-1}| = N$  then
4:      $S_i = F_{L-1}$ ;
5:   else
6:      $S_i = F_{L-1}$ ;
7:     for  $j = 1$  to  $N - |S|$  do
8:        $w_j \leftarrow W, j \in 1$  to  $N$ 
9:       for each  $s_k$  in  $F_L$  do
10:         $d_{\min} \leftarrow d(s_k)$ 
11:        for  $h = 2$  to end do
12:          if  $d(s_h) < d_{\min}$  then
13:             $d_{\min} \leftarrow d(s_h)$ 
14:             $w_j \leftarrow s_h$ 
15:          end if
16:        end for
17:      end for
18:       $\{s^1, s^2, \dots, s^H\} =$  Ascending order in  $w_i$  with PBI distance;
19:       $S_i \leftarrow s^1$ 
20:    end for
21:  end if
22: end for

```

However, in case (a), since $f(s_1(t), t)$ only slightly outperforms $f(s_2(t), t)$ in a very narrow time interval, it is entirely reasonable to regard $f(s_2(t), t)$ as the better fitness value. Therefore, the FNDS is defined as follows:

FNDS Definition: $s_i(x, t)$ dominates $s_j(x, t)$, denoted by $s_i(x, t) <_{\text{FNDS}} s_j(x, t)$ if and only if: (i) $f_m(s_i(x, t), t)$ is less than $f_m(s_j(x, t), t)$ over $[0, T]$ and $s_i(x, t) \neq s_j(x, t)$; or (ii) the integral of $f_m(s_i(x, t), t)$ is less than that of $f_m(s_j(x, t), t)$ over $[0, T]$, and $|f_m(s_i(x, t), t) - f_m(s_j(x, t), t)| < \epsilon, \epsilon > 0$, and $s_i(x, t) \neq s_j(x, t)$. Otherwise, it is denoted as $s_i(x, t) \parallel_{\text{FNDS}} s_j(x, t)$, that is, $s_i(x, t)$ and $s_j(x, t)$ do not dominate each other.

$$s_i(x, t) <_{\text{FNDS}} s_j(x, t) \Leftrightarrow \forall m \in \{1, 2, \dots, M\}, \quad (61)$$

$$\begin{cases} f_m(s_i(x, t), t) < f_m(s_j(x, t), t), s_i \neq s_j; \\ \int_0^T f_m(s_i(x, t), t) dt < \int_0^T f_m(s_j(x, t), t) dt \\ \wedge |f_m(s_i(x, t), t) - f_m(s_j(x, t), t)| < \epsilon, \epsilon > 0, s_i \neq s_j; \end{cases}$$

Next, we will demonstrate FNDS from the perspectives of irreflexive, asymmetric and transitive.

(i) If all values of $s(x, t)$ in $[0, T]$ are feasible solutions. Suppose that $s(x, t) <_{\text{FNDS}} s(x, t)$, for both of the two equivalent conditions in the FNDS definition, $s_i \neq s_j$; thus $s(x, t) <_{\text{FNDS}} s(x, t)$ is invalid, FNDS possesses irreflexive.

(ii) If all values of $s(x, t)$ in $[0, T]$ are feasible solutions, and it satisfies $s_i(x, t) <_{\text{FNDS}} s_j(x, t)$. Suppose that $s_j(x, t) <_{\text{FNDS}} s_i(x, t)$, according to the FNDS Definition, there are the following four situations:

- i) $f_m(s_i(x, t), t) < f_m(s_j(x, t), t)$ and $f_m(s_j(x, t), t) < f_m(s_i(x, t), t)$;
- ii) $f_m(s_i(x, t), t) < f_m(s_j(x, t), t)$ and the integral of $f_m(s_i(x, t), t)$ is less than that of $f_m(s_j(x, t), t)$;
- iii) the integral of $f_m(s_i(x, t), t)$ is less than that of $f_m(s_j(x, t), t)$ and $f_m(s_j(x, t), t) < f_m(s_i(x, t), t)$;
- vi) the integral of $f_m(s_i(x, t), t)$ is less than that of $f_m(s_j(x, t), t)$ and the integral of $f_m(s_j(x, t), t)$ is less than that of $f_m(s_i(x, t), t)$.

Obviously, i) and vi) are invalid. Regarding ii) and iii), as Eq. (62).

$$f_m(s_i(x, t), t) < f_m(s_j(x, t), t) \Rightarrow \int_0^T f_m(s_i(x, t), t) dt < \int_0^T f_m(s_j(x, t), t) dt \quad (62)$$

It is because the B-spline function is continuous on the interval $[0, T]$. Thus, $s_j(x, t) <_{\text{FNDS}} s_i(x, t)$ is invalid, FNDS possesses asymmetric.

(iii) If $s_i(x, t)$, $s_j(x, t)$ and $s_k(x, t)$ are all feasible solutions, satisfy $s_i(x, t) <_{\text{FNDS}} s_j(x, t)$, $s_j(x, t) <_{\text{FNDS}} s_k(x, t)$, then $f_m(s_i(x, t), t) < f_m(s_j(x, t), t)$ and $f_m(s_j(x, t), t) < f_m(s_k(x, t), t)$, thus $f_m(s_i(x, t), t) < f_m(s_k(x, t), t)$, $s_i(x, t) <_{\text{FNDS}} s_k(x, t)$, FNDS possesses transitive.

Considering constrained CFNDS, which follows the same procedure as the constrained Pareto dominated (CDP) process, as shown in Eq. (63).

$$s_i(x, t) <_{\text{CFNDS}} s_j(x, t) \Leftrightarrow \begin{cases} V(s_i(x, t)) = 0 \wedge V(s_j(x, t)) > 0; \\ V(s_i(x, t)) = V(s_j(x, t)) \wedge s_i(x, t) <_{\text{FNDS}} s_j(x, t); \\ V(s_i(x, t)) < V(s_j(x, t)); \\ V = \sum_1^M v_m + \sum_0^T v_t \end{cases} \quad (63)$$

Here, v_m denotes the constraint violation magnitude for each objective, and v_t denotes the constraint violation magnitude at each time step.

Upon completion of the CFNDS, if the number of individuals in the $(L - 1)$ th front is equal to the desired population size N , all individuals in that front are directly selected. Otherwise, individuals from the last front F_L are iteratively selected to fill the population.

During this truncation process, each candidate individual is assigned to the corresponding reference vector w_i , allowing for multiple assignments. Then, for each subset within a cluster, the Penalty Boundary Intersection (PBI) distance is computed to identify the most suitable individuals that balance convergence and diversity. Individuals with the smallest PBI distances are selected sequentially until the population reaches the desired size N . The computation of PBI follows the same format as that presented in Ming et al. (2023). Finally, from each cluster, the individual with the smallest distance is selected.

4.2.3. Computational complexity analysis

In the proposed PC-MAOEA-PINN algorithm, the environmental selection procedure is applied to two combined populations, CS_1 and CS_2 , in each generation. For each population, individuals are first ranked using the CFNDS non-dominated sorting criterion, producing a series of non-dominated fronts $F = [F_1, F_2, \dots, F_L]$. Assuming a population size of N and M objectives, the complexity of CFNDS sorting is $O(MN^2)$, as it extends classical non-dominated sorting by incorporating additional scheduling time considerations.

After sorting, a truncation procedure is applied to the last front F_L to fill the population to size N . For each candidate individual, the corresponding reference vector w_i is considered, and the Penalty Boundary Intersection (PBI) distance is computed to select the most suitable individual. Unlike pairwise comparisons among all individuals, this procedure randomly selects candidates and updates the current minimum distance if a smaller value is found, which is conceptually similar to the update step in the K-means algorithm. Consequently, in the worst

case, where the last front contains $|F_L| \sim N$ individuals, the truncation process has a complexity of $O(N)$. Therefore, the overall complexity of a single environmental selection operation is dominated by the CFNDS ranking, yielding $O(N^2)$.

The PCEO operator is employed to generate offspring by adaptively sampling from the probabilistic distribution of each gene based on Shapley values. For a population of size N , with decision variables of dimension D and K groups, the operator iterates over all individuals and decision variables. Within each iteration, the Shapley value of each group is computed, followed by the calculation of the mean and variance of the gene, and a sample is drawn from the corresponding normal distribution. Assuming that the computation of each Shapley value is of constant complexity $O(1)$ (or bounded by a small group size), the overall time complexity of the PCEO operator is dominated by the three nested loops over individuals, genes, and groups. Therefore, the complexity can be expressed as $O(N \cdot D \cdot K)$. When treating D and K as constants, this reduces to $O(N)$ per individual and $O(N^2)$ for the entire population.

In summary, the overall computational complexity of the PC-MAOEA-PINN algorithm is $O(N^2)$. Here, the complexity of evaluating the pre-trained PINNSM surrogate model $\mathcal{H}$ is not considered, as it is assumed to be available in advance.

5. Experiments and analysis

This subsection presents the validation of the proposed model and algorithms. The base model employed in this study is the MATPOWER 39-bus test system (Murillo-Sánchez et al., 2013; Wang et al., 2007; Zimmerman et al., 2010). We have extended this benchmark power system into an IES that includes power, heat, gas, and cooling subsystems. The original MATPOWER 39-bus system comprises 10 PVs, 38 PQs, and 1 Slack node.

To enhance the multi-energy characteristics of IES, we modified bus 30 to incorporate a CHP, while buses 33 and 37 were adapted to host GTs. Buses 38 and 39 were equipped with WIN and PH, respectively, to represent renewable energy integration. Additionally, a heat subsystem, a gas subsystem, and a cooling subsystem were integrated into the IES.

The heat subsystem consists of 2 HSs (CHP and an EB), 3 heat load nodes (HDS), and 1 Slack node. The EB is from node 24 in power subsystem and connected to node 3 in heat subsystem. The CHP is from node 30 in power subsystem and connected to node 1 in heat subsystem.

The cooling subsystem mirrors the structure of the thermal subsystem, with two cooling sources located at buses 33 and 37 and connected to node 1 and nodes 3, respectively. The gas subsystem adopts the Belgian 20-node natural gas network (He et al., 2025), which includes 6 GSs and 20 GDs. Using real-world RES and load data from the electricity market, and following the scenario reduction method described in Almeida et al. (2024), 9 representative scenarios were generated and proportionally allocated to the corresponding nodes in the IES model based on the installed capacity and load ratios. The key model parameters are summarized in Table 3 and detailed data for the remaining system parameters can be found in Appendix A.

5.1. Contrast experience

In this subsection, we compare the proposed algorithm with four state-of-the-art evolutionary algorithms: NSGA-III (Deb & Jain, 2013), FLEA (Li et al., 2022), CMME (Ming et al., 2022), and thetaDEA-CPBI (Ming et al., 2023), across all 16 scheduling scenarios. For each algorithm, a dedicated PINNSM was trained and employed during the optimization process to estimate the CVD for each individual. Notably, For the NSGA-III, we adopted its constrained version, i.e., the Pareto non-dominated sorting with the CVD strategy, to handle constraints. For the FLEA, since it is a derivative under the MOEA framework, we employed a penalty-based approach for constraint handling. For the CMME and theta-DEA-CPBI, as they were originally designed for constrained problems, we directly applied them without modification.

Table 3
Model parameters.

Parameters	Values	Parameters	Values
η_{g-h}	0.50	met	1.1
η_{g-e}	0.35	N_T	71
η_{CHP}	0.85	N_{ELE}	39
η_{EB}	0.95	N_{BESS}	1
η_{dis}	0.95	$N_{ELE,LOAD}$	38
η_{ch}	0.95	N_F	5
η_{ABS}	0.70	N_{EBR}	29
η_{HRS}	[0.75, 0.8]	N_{PV+WIN}	2
η_{GT}	[0.35, 0.4]	N_{GAS}	20
c_p	1.167×10^{-6}	N_{GT}	2
Q_{LHV}	35.17	N_{EB}	1
c/o	0.23–2.0	N_{CHP}	1

To evaluate performance, two widely used many-objective optimization metrics were adopted: Hypervolume (HV) and Inverted Generational Distance Plus (IGD+). All objective values were normalized to the range [0, 1]. The HV metric measures the volume between the obtained Pareto front and a reference point (typically at 1.1), with larger values indicating better convergence and diversity.

The IGD+ metric, on the other hand, calculates the average distance from the obtained solutions to a predefined Pareto front. Although the reference front may not represent the true optimal front, the IGD+ values still provide a useful comparative reference. The results for HV and IGD+ are summarized in Tables 4 and 5, respectively.

For each test case, the mean and standard deviation of both HV and IGD+ indicators are calculated over 10 independent runs. Statistical significance is assessed based on pairwise comparisons with the proposed PC-MaOEA-PINN. The symbols “+”, “-”, and “=” indicate that a competing algorithm performs significantly better than, significantly worse than, or statistically equivalent to the proposed method in the rank sum test, respectively. The best mean value among the five algorithms for each case is highlighted in **bold**.

From the results in Table 4, it can be observed that PC-MaOEA-PINN demonstrates superior and consistent performance across all test cases. Specifically, PC-MaOEA-PINN achieves the highest or statistically equiv-

alent HV values in all nine cases, indicating robust convergence and diversity performance.

Among the compared algorithms, CMME-PINN shows the closest performance, with three cases statistically equivalent and one case significantly superior to the proposed method. In contrast, FLEA-PINN, NSGAIII-PINN, and θ -DEA-CPBI-PINN perform the worst in terms of HV, with eight out of nine cases being significantly inferior, respectively. The comparison highlights that PC-MaOEA-PINN maintains a clear advantage in HV performance while ensuring solution stability, as reflected by lower standard deviations in most scenarios.

From the results in Table 5, it is evident that PC-MaOEA-PINN achieves the best or statistically equivalent IGD+ performance across all test cases, reflecting its strong convergence ability and approximation quality to the Pareto front. Notably, in Cases 4, 7, and 8, the proposed algorithm outperforms all baselines with statistical significance, achieving the lowest IGD+ values. In addition, PC-MaOEA-PINN achieves the best mean IGD+ value in seven out of nine test cases, further confirming its superiority in convergence accuracy across a wide range of problem instances.

Fig. 9 presents the convergence curves of the five algorithms in terms of the HV metric, which reflects the diversity and convergence quality of the obtained solution sets. The shaded areas represent the standard deviations over multiple independent runs, indicating the robustness of each algorithm.

As shown in the figure, PC-MaOEA-PINN demonstrates a significantly faster convergence rate in most cases, typically approaching optimal HV values within 20 to 40 iterations, which is notably faster than the other algorithms (e.g., NSGAIII-PINN and FLEA-PINN), many of which require over 60 iterations to stabilize.

Moreover, PC-MaOEA-PINN exhibits narrower confidence intervals in several scenarios, suggesting higher robustness and lower performance variance. Although CMME-PINN slightly outperforms PC-MaOEA-PINN in a few instances (such as Cases b and i), the overall performance of PC-MaOEA-PINN remains superior across most test cases.

Fig. 10 shows the convergence behavior of the five algorithms with respect to the IGD metric, which measures the distance between the obtained solution set and the true Pareto front.

Table 4
Comparison of HV (Mean $\pm$ STD) across cases 1–9 for five algorithms.

Problem	NSGAIII-PINN	FLEA-PINN	CMME-PINN	θ -DEA-CPBI-PINN	PC-MaOEA-PINN
Case_1	6.4433e-1(1.49e-1)-	4.7819e-1(6.39e-2)-	7.0138e-1(1.39e-1) =	6.7251e-1 (1.53e-1)-	7.4734e-1 (1.13e-1)
Case_2	6.8730e-1(1.64e-1)-	7.6676e-1 (1.81e-1) =	6.7795e-1(1.46e-1)-	6.5164e-1(1.80e-1)-	7.6050e-1(8.46e-2)
Case_3	6.0542e-1(1.17e-1)-	4.8590e-1(1.47e-1)-	6.4073e-1 (1.22e-1)-	6.7670e-1(1.02e-1)-	7.3807e-1 (1.15e-1)
Case_4	7.1387e-1(1.90e-1)-	5.2184e-1(1.00e-1)-	7.1646e-1(1.29e-1)-	7.6164e-1(1.05e-1)-	8.1418e-1 (1.11e-1)
Case_5	5.6602e-1(1.64e-1)-	5.6284e-1(1.78e-1)-	6.4662e-1(1.37e-1)-	5.9821e-1(1.38e-1)-	7.1097e-1 (1.39e-1)
Case_6	6.3157e-1(1.50e-1)-	4.7398e-1(1.33e-1)-	7.0332e-1(1.05e-1) =	6.9046e-1(1.23e-1)-	7.5031e-1 (1.62e-1)
Case_7	5.8007e-1(1.50e-1)-	5.7934e-1(1.92e-1)-	7.1866e-1(8.85e-2) =	7.0156e-1(1.24e-1)-	7.5608e-1 (8.56e-2)
Case_8	6.7125e-1(1.77e-1)-	4.8613e-1(1.74e-1)-	6.6498e-1(1.14e-1)-	6.0195e-1(1.58e-1)-	8.2783e-1 (1.12e-1)
Case_9	6.4716e-1(1.24e-1) =	4.5816e-1(1.02e-1)-	7.1425e-1 (1.10e-1) +	6.4070e-1(1.60e-1) =	6.4782e-1(1.04e-1)
+ / - / =	0 / 8 / 1	0 / 8 / 1	1 / 5 / 3	0 / 8 / 1	-

Table 5
Comparison of IGD (Mean $\pm$ STD) across cases 1–9 for five algorithms.

Problem	NSGAIII-PINN	FLEA-PINN	CMME-PINN	θ -DEA-CPBI-PINN	PC-MaOEA-PINN
Case_1	1.4932e-1 (7.42e-2) =	2.3627e-1 (4.78e-2)-	1.3371e-1 (6.89e-2) =	1.3905e-1 (6.95e-2) =	1.1603e-1 (5.37e-2)
Case_2	1.3289e-1 (8.92e-2) =	9.3398e-2 (7.34e-2) +	1.4781e-1 (7.92e-2) =	1.5090e-1 (1.06e-1) =	1.0658e-1 (4.35e-2)
Case_3	1.7389e-1 (6.25e-2)-	2.3612e-1 (1.02e-1)-	1.6223e-1 (6.57e-2) =	1.3658e-1 (5.18e-2) =	1.1934e-1 (5.90e-2)
Case_4	1.2309e-1 (9.33e-2)-	2.0982e-1 (6.33e-2)-	1.2764e-1 (6.56e-2)-	9.4690e-2 (4.61e-2) =	8.1684e-2 (5.33e-2)
Case_5	2.0185e-1 (9.50e-2)-	1.9842e-1 (1.00e-1)-	1.5945e-1 (7.29e-2) =	1.8402e-1 (8.03e-2) =	1.3437e-1 (6.52e-2)
Case_6	1.7337e-1 (8.13e-2)-	2.5240e-1 (9.18e-2)-	1.3168e-1 (4.66e-2) =	1.3228e-1 (6.26e-2) =	1.0989e-1 (8.06e-2)
Case_7	1.8989e-1 (8.24e-2)-	1.8902e-1 (9.70e-2)-	1.2236e-1 (4.11e-2)-	1.3270e-1 (6.43e-2)-	9.7771e-2 (4.00e-2)
Case_8	1.4402e-1 (9.39e-2)-	2.5027e-1 (9.95e-2)-	1.5069e-1 (5.77e-2)-	1.8391e-1 (9.13e-2)-	7.9105e-2 (5.24e-2)
Case_9	1.5042e-1 (7.17e-2) =	2.5656e-1 (6.54e-2)-	1.2201e-1 (5.65e-2) =	1.5886e-1 (9.13e-2) =	1.6378e-1 (5.32e-2)
+ / - / =	0 / 6 / 3	1 / 8 / 0	0 / 3 / 6	0 / 2 / 7	-

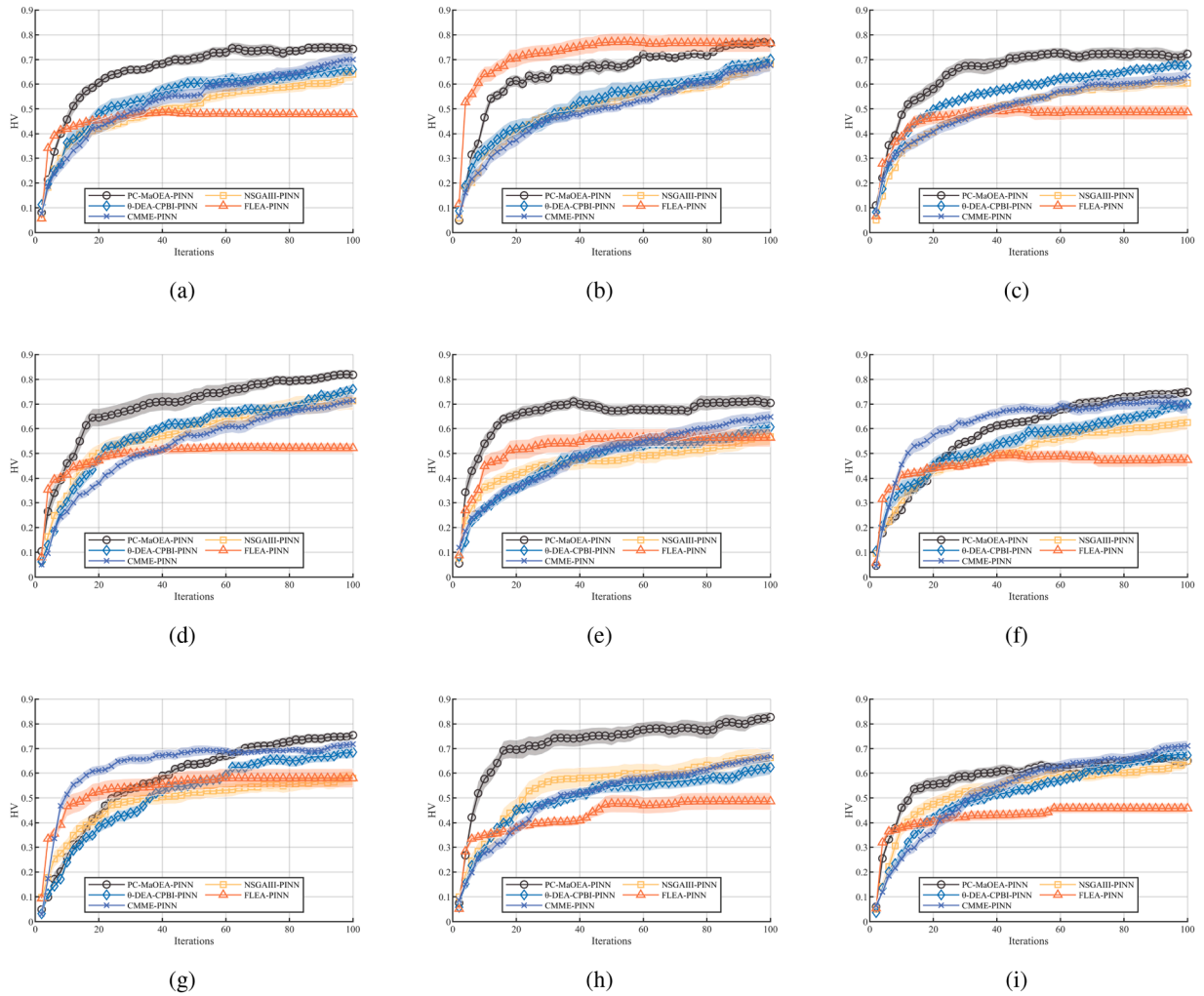


Fig. 9. The HV convergence trend on five algorithms. (a)-(i): case1-csae9.

It can be observed that PC-MaOEA-PINN consistently exhibits faster and more stable convergence across all test cases. In most scenarios, it significantly reduces the IGD value within the first 20 to 30 iterations and maintains a low level afterward, outperforming the other algorithms.

Particularly, the IGD curves of PC-MaOEA-PINN are generally located at the bottom, indicating better proximity to the true front. Furthermore, the narrow-shaded areas reflect high consistency among runs, verifying the robustness of the method.

In summary, PC-MaOEA-PINN achieves superior convergence speed, solution quality, and stability, highlighting its effectiveness in solving multi-objective optimization problems

5.2. Statistical experiment

To rigorously validate the performance superiority of the proposed PC-MaOEA-PINN over the four baseline algorithms (NSGAIII-PINN, FLEA-PINN, CMME-PINN, and θ -DEA-CPBI-PINN), three types of statistical tests were conducted: the Friedman test, the Wilcoxon signed-rank test, and the Bayesian sign test. These tests were applied to the HV and the IGD + indicators across Case1-Case9.

Table 6 presents the p-values obtained from the Friedman and Wilcoxon tests. For HV, the Friedman test reports p-values less than 0.05 for all pairwise comparisons, indicating significant performance differences across algorithms. The Wilcoxon test further confirms these results, with p-values also below 0.05 in most cases (except for FLEA-

Table 6

The P-value of statistical tests on case 1–9.

Metric	Test	NSGAIII-PINN	FLEA-PINN	CMME-PINN	θ -DEA-CPBI-PINN
HV	Friedman	0.0027	0.0196	0.0196	0.0027
	Wilcoxon	0.0039	0.0078	0.0391	0.0039
IGD +	Friedman	0.0196	0.0196	0.0196	0.0196
	Wilcoxon	0.0078	0.0078	0.0547	0.0078

$p < .05$ indicates rejection of the null hypothesis

PINN on IGD + with $p = 0.0547$, slightly above the threshold). These results reject the null hypothesis of equal performance and suggest that PC-MaOEA-PINN significantly outperforms the other algorithms in most scenarios.

To further analyze, we employed the Bayesian sign test, as shown in Table 7. In our Bayesian tests, we set the region of practical equivalence (ROPE) to 0.05 for HV and 0.01 for IGD +, reflecting a moderately strict criterion for practical indistinguishability. These settings are justified by the scale and sensitivity of the two indicators: HV values are generally larger and tolerate minor fluctuations, while IGD + is highly sensitive to small deviations near zero.

Fig. 11 presents the probability distribution and the Bayesian triangle plot. Under these settings, the proposed PC-MaOEA-PINN demonstrates dominant performance in nearly all comparisons, with negligible ROPE probabilities in most cases. For the case of θ -DEA-CPBI in IGD +,

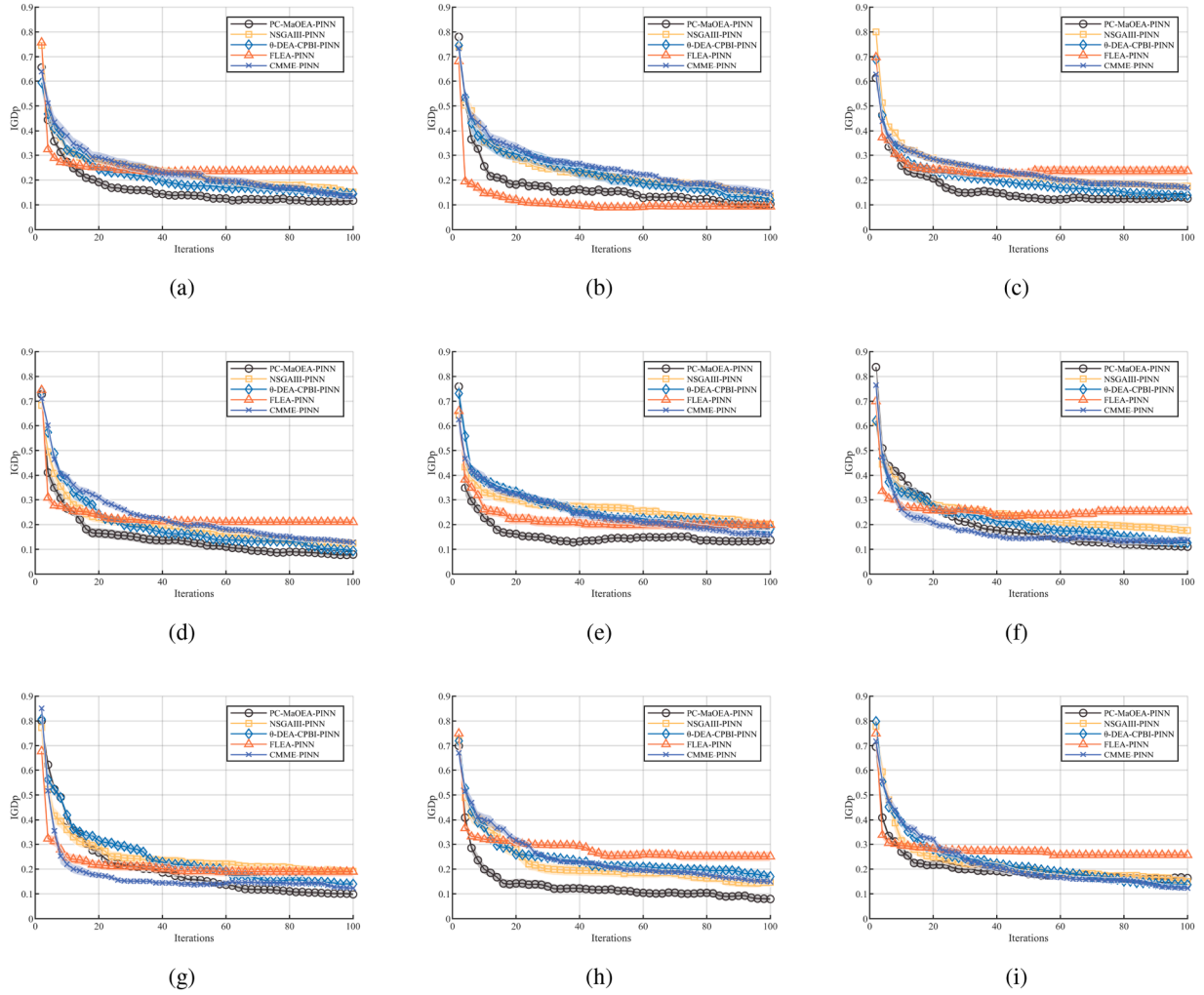


Fig. 10. The IGD+ convergence trend on five algorithms. (a)-(i): case1-csae9.

Table 7

The probability of dominance obtained in Bayesian tests.

Metric	Region	NSGAIII-PINN	FLEA-PINN	CMME-PINN	θ -DEA-CPBI-PINN
HV	Left	0.000	0.000	0.002	0.000
	Rope	0.016	0.024	24.372	3.058
	Right	99.984	99.976	75.626	96.942
<i>Rope</i> = 0.05, <i>prior</i> = 0.5					
IGD+	Left	0.018	0.006	0.346	0.000
	Rope	0.004	0.008	0.788	0.956
	Right	99.978	99.986	98.866	99.044
<i>Rope</i> = 0.01, <i>prior</i> = 0.5					

a relatively high ROPE probability (95.6 %) suggests practical equivalence, aligning with our chosen threshold.

The results reveal that, for the HV metric, the dominance probability of PC-MaOEA-PINN consistently exceeds 96 % across all cases, clearly indicating its overwhelming performance advantage. The Right-probabilities—which quantify the likelihood that PC-MaOEA-PINN outperforms its counterparts—reach 99.984 %, 99.976 %, 75.626 %, and 96.942 % when compared to NSGAIII-PINN, FLEA-PINN, CMME-PINN, and θ -DEA-CPBI-PINN, respectively.

It is particularly notable that the ROPE (Region of Practical Equivalence) probabilities are nearly zero in most cases, further confirming that the observed differences are not only statistically significant but also

practically meaningful. This suggests that PC-MaOEA-PINN is consistently more effective in achieving better hypervolume coverage, which reflects improved convergence toward the Pareto front as well as enhanced solution diversity.

A similar trend is observed in the IGD+ comparisons. The dominance probability of PC-MaOEA-PINN is above 98.8 % in all comparisons, with the only slight exception being the comparison against FLEA-PINN, where the Right-probability is 98.866 % and the ROPE probability reaches 0.788 %. Although this marginal increase in ROPE indicates a minor uncertainty in practical equivalence, the probability mass remains clearly skewed in favor of PC-MaOEA-PINN, affirming its comparative advantage.

Overall, the consistent results across all three types of tests provide robust evidence that PC-MaOEA-PINN outperforms baseline methods in both convergence quality and solution diversity, demonstrating its effectiveness and reliability.

In addition, Figs. 12–14 present the box plots of the objective values obtained by the five algorithms over multiple independent runs. Specifically, Fig. 12 corresponds to Cases 1–3, Fig. 13 to Cases 4–6, and Fig. 14 to Cases 7–9. Since all objectives in high-dimensional multi-objective optimization are uniformly transformed into minimization problems, smaller mean values and quartiles in the box plots consistently indicate better performance.

In Case 1, it can be observed that PC-MaOEA achieves the lowest mean across all five objectives. Moreover, for the objectives of profit, RES, and stability, the IQR ranges are the smallest, suggesting that the

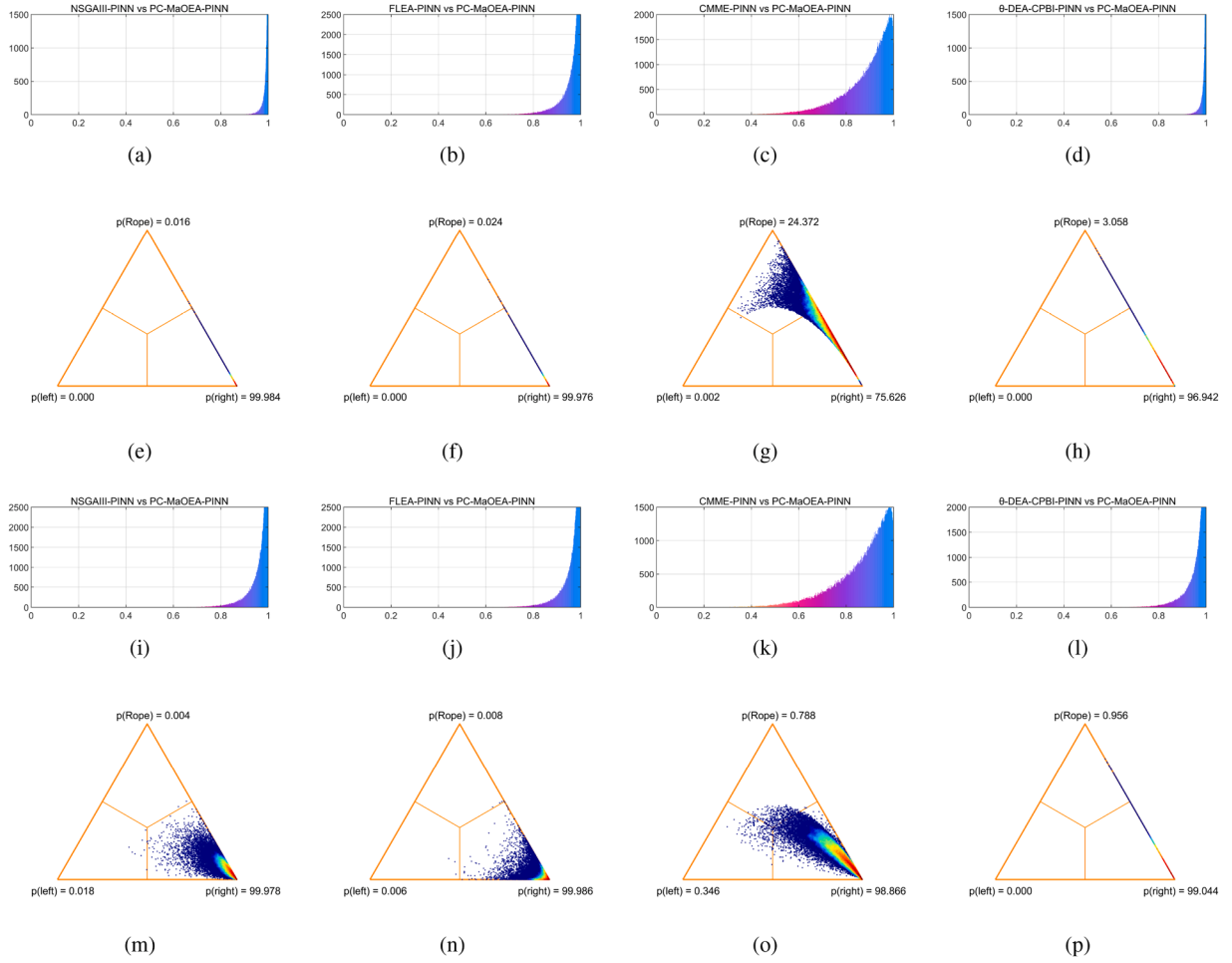


Fig. 11. The result of the Bayesian sign test. (a) - (d): The probability distribution of the HV indicator. (e)-(h): Bayesian triangle plot corresponds to (a) - (d). (i)-(l): The probability distribution of the IGD+ indicator. (m)-(p): Bayesian triangle plot corresponds to (i) - (l).

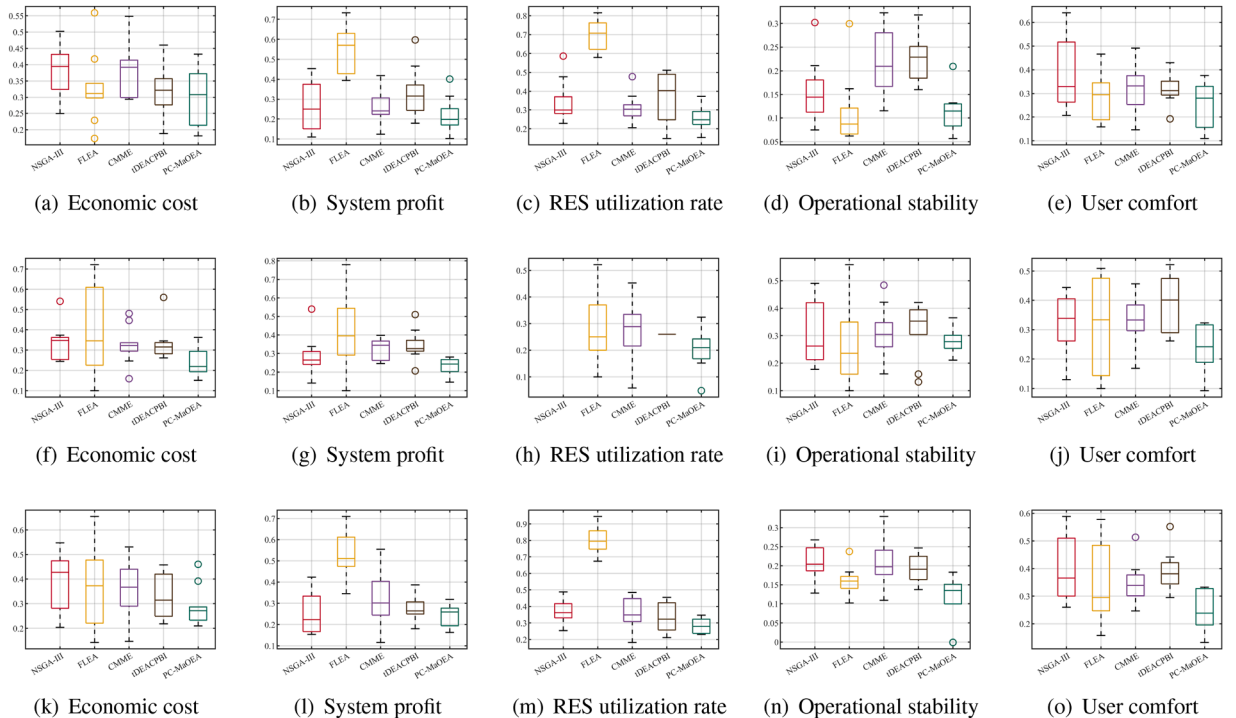


Fig. 12. The result of box plots in Case 1–3. (a) - (e): Case 1. (f)-(j): Case 2. (k)-(o): Case 3.

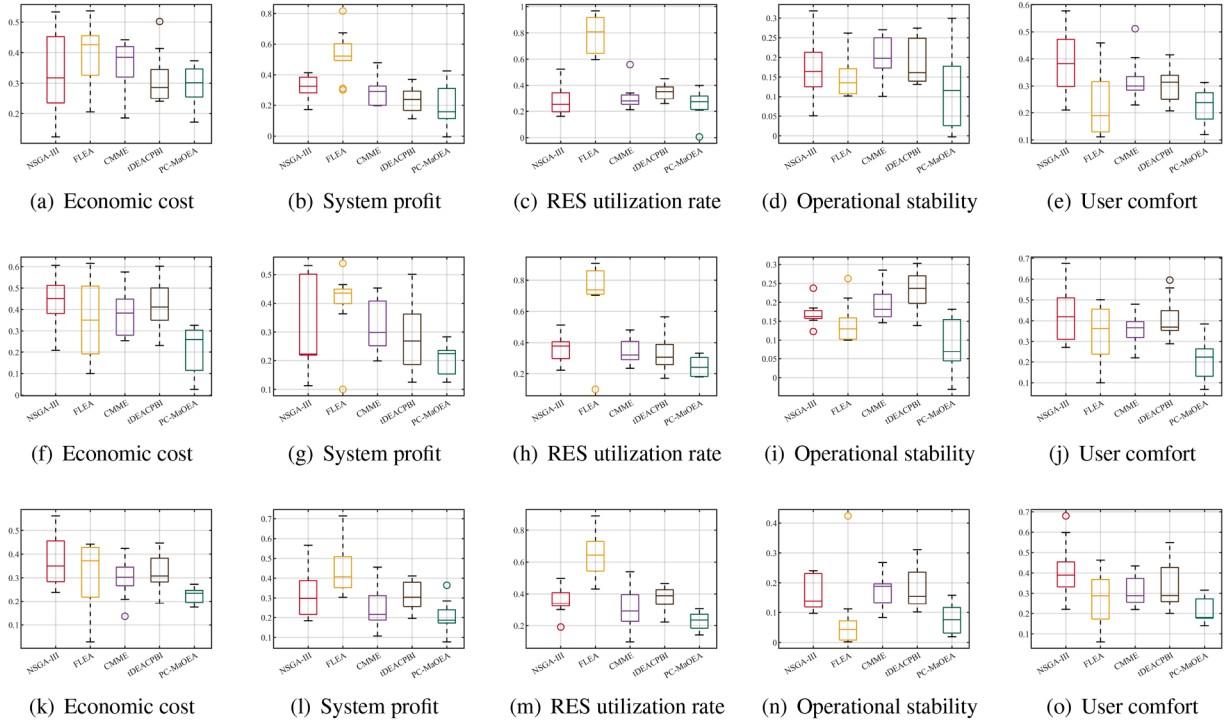


Fig. 13. The result of box plots in Case 4–6. (a) - (e): Case 4. (f)-(j): Case 5. (k)-(o): Case 6.

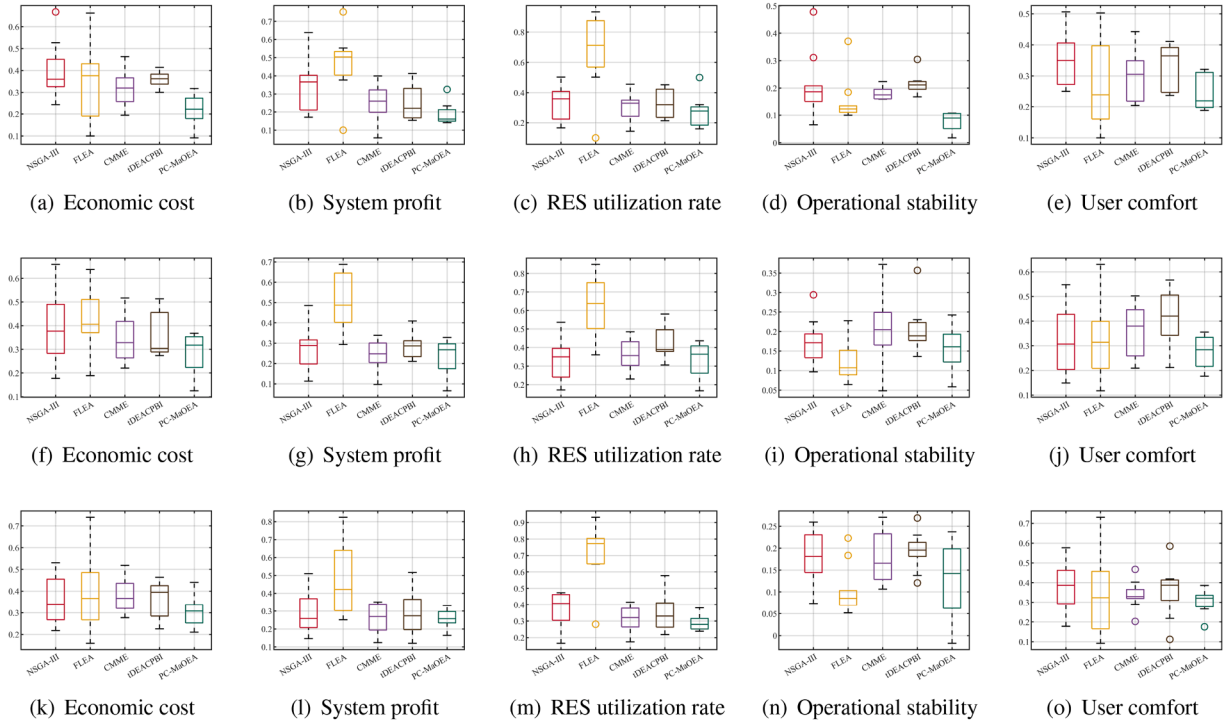


Fig. 14. The result of box plots in Case 7–9. (a) - (e): Case 7. (f)-(j): Case 8. (k)-(o): Case 9.

algorithm exhibits higher stability. In Case 2, our proposed PC algorithm again achieves the best performance in terms of mean values. Although its lower quartile is slightly inferior to that of FLEA for certain objectives, FLEA suffers from poor robustness, as reflected by its considerably wider box spans.

In Case 3, the performance gaps among the algorithms are narrower. Nevertheless, PC-MaOEA still demonstrates superior performance in both mean and dispersion: it consistently yields lower mean values in

most objectives while maintaining relatively compact IQR ranges. This indicates that even in scenarios where the optimization difficulty increases and algorithmic performance tends to converge, PC-MaOEA is capable of providing more stable and reliable solutions compared to the other competitors.

In Case 4, PC-MaOEA continues to exhibit strong performance, where its mean values remain among the lowest across most objectives. More importantly, the box plots reveal much narrower IQR ranges compared

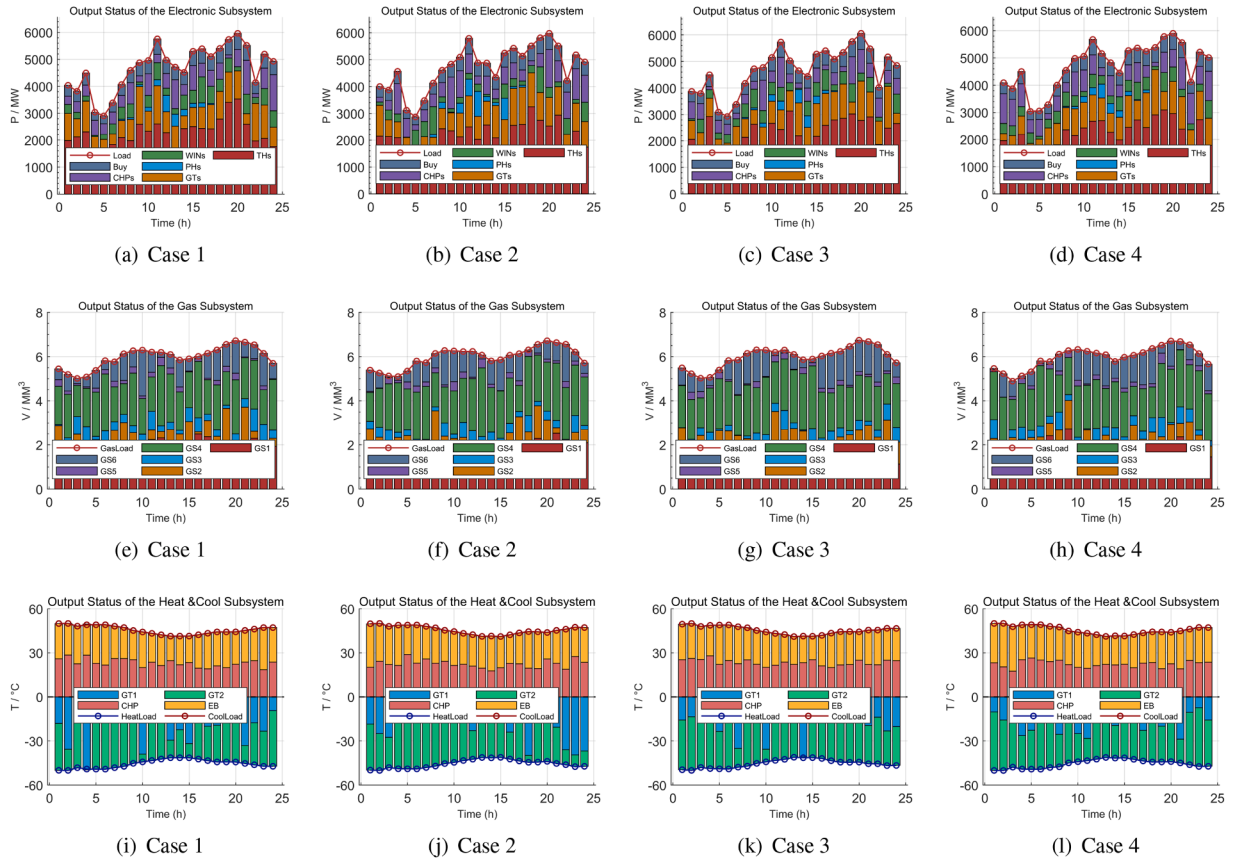


Fig. 15. The Output status of the electronic subsystem, the gas subsystem, and the heat & cool subsystem in Case 1–4.

to the other algorithms, further confirming its robustness. In contrast, algorithms such as NSGAIPI-PINN and FLEA show wider dispersions, indicating higher variability and weaker reliability.

In Case 5, the advantage of PC-MaOEA becomes even more evident. The algorithm not only maintains the lowest mean values but also exhibits highly compact box spans in all five objectives. This highlights its ability to achieve both accuracy and robustness simultaneously, a property that competing algorithms fail to deliver consistently.

In Case 6, while the differences in mean performance between algorithms are less pronounced, PC-MaOEA still achieves relatively stable outcomes. Although some competing algorithms occasionally attain competitive mean values, their wider IQR ranges suggest inferior robustness. PC-MaOEA strikes a more favorable balance between convergence accuracy and solution stability.

In Case 7, PC-MaOEA once again outperforms the other algorithms by achieving lower means across most objectives. The narrow IQR ranges further emphasize its consistency, while other algorithms exhibit significantly larger box plots, implying greater uncertainty in their optimization outcomes.

In Case 8, the performance differences become sharper. PC-MaOEA not only achieves superior mean values but also demonstrates the smallest box spans across nearly all objectives. By contrast, FLEA and NSGAIPI-PINN show highly fluctuating results, which indicates weaker robustness in handling the problem complexity.

Finally, in Case 9, PC-MaOEA continues to maintain its advantage. Despite the increased difficulty of the scenario, it delivers relatively lower means and stable IQR ranges, underscoring its capability to adapt to more challenging optimization tasks. The competing algorithms, in contrast, suffer from larger dispersion and less reliable performance.

Taken together, the results from Cases 1–9 consistently demonstrate that PC-MaOEA not only achieves competitive or superior objective val-

ues but also provides significantly improved stability and robustness compared to state-of-the-art algorithms.

5.3. Performance experience

In this subsection, the performance of the proposed multi-energy system optimization model and algorithm is evaluated under Case 1–9. Fig. 15(a)–(d) illustrate the output profiles of the electricity, (e)–(h) the gas, and (i)–(l) the heating & cooling subsystems in Case 1–4, over the scheduling horizon, revealing the characteristics of energy flow coordination and system-level operational behavior.

In the electricity subsystem, THs provide the majority of the power supply, maintaining a relatively high and stable output throughout the scheduling period. This indicates their dominant role and cost-effectiveness in the overall system. GTs and CHPs rank next, primarily serving as flexible resources to handle load fluctuations. PHs exhibit typical intermittency and solar dependency, generating electricity only during daytime hours. In contrast, WNs operate continuously throughout the day with relatively stable output, despite some variability, thereby contributing significantly to the utilization of renewable energy resources.

In the gas subsystem, among the six gas sources, GS1 and GS4 exhibit the highest output. This is mainly due to their central positioning within the gas network and their connections to multiple gas load nodes. Specifically, GS1 is directly connected to CHP units, supplying gas for both electricity and heat generation, making it a key node in the system. This distribution aligns well with practical infrastructure and gas flow characteristics.

In the heating subsystem, the outputs of CHPs and EBs are approximately equal. Since the heating subsystem comprises only six nodes with relatively low thermal demand, both EBs (electric load side) and CHPs

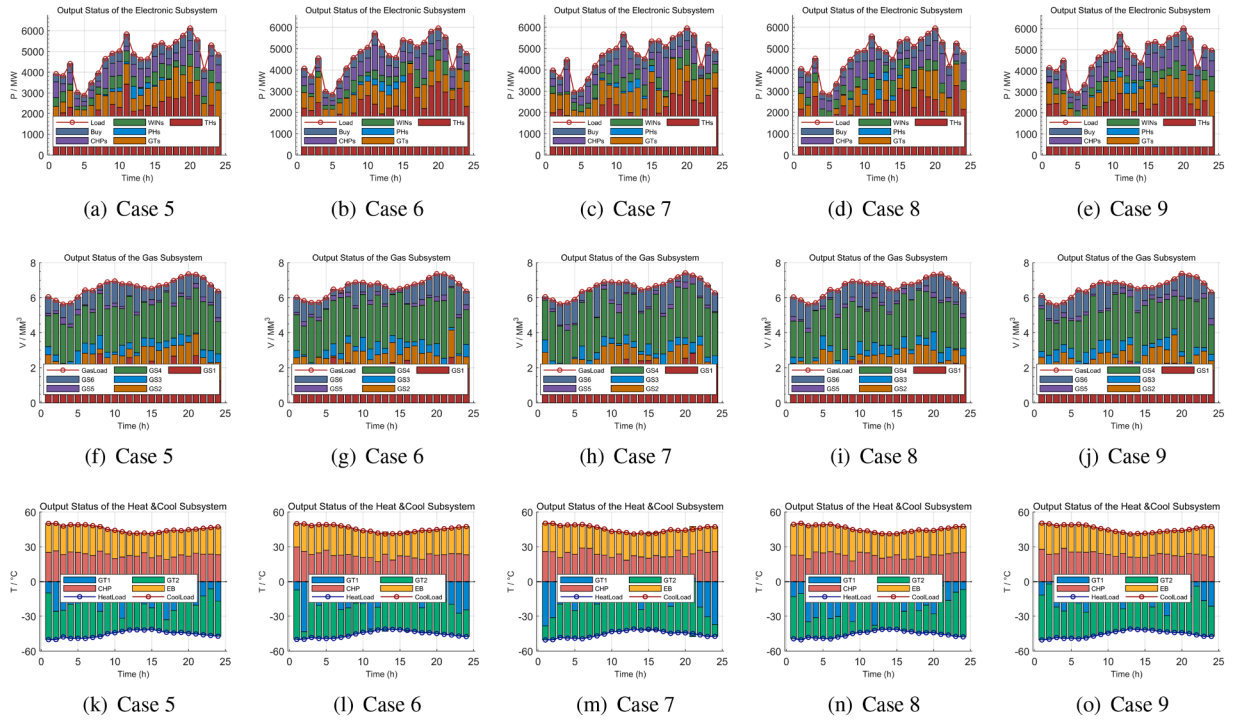


Fig. 16. The Output status of the electronic subsystem, the gas subsystem, and the heat & cool subsystem in Case 5–9.

(gas load side) share the heating burden. This reflects the system's flexible dispatch strategy under light-load conditions. Notably, CHPs serve as tri-generation units, participating in electricity, heat, and gas coupling, while EBs offer fast response capabilities, highlighting the complementary nature of different energy carriers.

In the cooling subsystem, the cooling supply is provided via ABSs connected to two gas turbines, GT1 and GT2. The results indicate that GT1 contributes more cooling output than GT2, primarily due to its higher rated power capacity. Consequently, the waste heat from GT1 allows greater thermal-to-cooling conversion via ABS. This demonstrates the model's capability to effectively utilize excess heat for cooling, improving overall energy efficiency and validating the coordination strategy between gas and cooling energy flows.

Figs. 16(a)–(e) illustrate the output profiles of the electricity, (f)–(j) the gas, and (k)–(o) the heating & cooling subsystems in Case 5–9, over the scheduling horizon. The results further reveal the characteristics of energy flow coordination and system-level operational behavior under more complex conditions.

In the electricity subsystem, a greater contribution from renewable energy sources is observed compared with earlier cases, particularly from PHs and WINs. The proposed model effectively accommodates the intermittency of PHs and the variability of WINs by dynamically coordinating THs and CHPs, thereby ensuring a reliable balance between supply and demand. This indicates the system's ability to integrate renewable resources while maintaining operational stability.

In the gas subsystem, the outputs become more diversified across the six gas sources. While GS1 and GS4 remain dominant, GS2 and GS5 exhibit increased participation due to shifting load patterns and network pressure balancing requirements. This reflects the adaptability of the gas network in redistributing flows under varying system conditions, while ensuring a steady supply to CHP units and gas load nodes.

In the heating subsystem, rising demand levels cause CHPs to play a more significant role in supplying thermal energy, supported by additional input from EBs. The joint participation of CHPs and EBs underlines the system's capacity to balance gas- and electricity-driven heating, demonstrating flexible energy carrier substitution and enhanced operational resilience under varying load conditions.

In the cooling subsystem, ABS units continue to leverage the waste heat from GTs, but their contributions become more balanced between GT1 and GT2 as demand increases. This indicates that the coordination strategy allows for efficient allocation of thermal-to-cooling conversion capacity, avoiding over-reliance on a single source and improving cooling reliability in high-demand scenarios.

In summary, the experimental results across Cases 1–9 validate the feasibility and effectiveness of the proposed model and algorithm for multi-energy system scheduling under multi-temporal and multi-energy coordination scenarios. The method demonstrates a strong capability to coordinate resources across subsystems, achieving a balanced trade-off among economic efficiency, environmental sustainability, and operational stability.

Furthermore, to evaluate the performance of the algorithm in solving many-objective optimization problems, we analyzed the convergence trends of the five objective functions throughout the iterative process, as illustrated in Fig. 17 in Case 1.

The Economic cost exhibits a pronounced downward trend, decreasing by approximately 0.3×10^7 USD, which reflects the effectiveness of the proposed optimization framework in reducing overall operational expenses. In parallel, the System profit increases by about 1.2×10^5 USD, primarily driven by strategic participation in the carbon market, demonstrating that the integration of economic incentives can simultaneously enhance revenue while promoting low-carbon operation.

The observed economic improvements are closely associated with an increased utilization rate of renewable energy sources, particularly photovoltaic and wind power. This trend indicates that the optimization not only prioritizes cost reduction and profit maximization but also facilitates more efficient integration of clean energy, contributing to sustainable system operation. At the same time, the system stability indicator gradually declines, suggesting a reduction in power output fluctuations across subsystems. This improvement in operational stability ensures more robust and reliable system performance, which is critical for maintaining safe and continuous energy supply under variable renewable generation and load conditions.

Regarding user comfort, thermal load node 4, representing an indoor office environment during summer, demonstrates RMV values

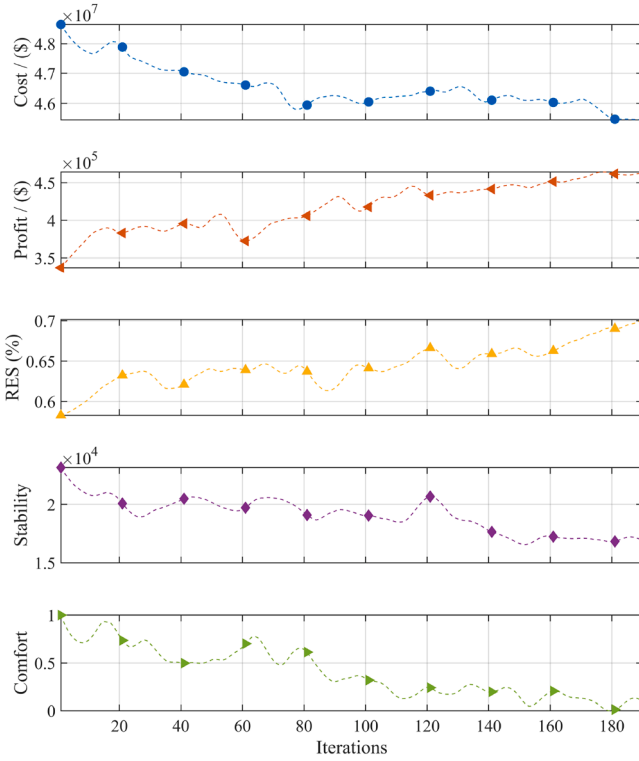


Fig. 17. The iteration plot of objective value in Case 1.

converging toward 0.5. This convergence indicates that the indoor thermal environment remains within an acceptable comfort range, confirming that the proposed scheduling approach can effectively balance energy efficiency with user satisfaction.

Overall, the results demonstrate that the proposed optimization framework achieves a synergistic improvement across multiple objectives. By simultaneously reducing economic costs, increasing system profits, enhancing renewable energy utilization, stabilizing operational performance, and maintaining user comfort, the framework provides a comprehensive solution for coordinated multi-objective energy management. These findings underscore the potential of integrating economic, environmental, and user-centric considerations in the operation of IES.

6. Conclusion

This paper investigates the many-objective scheduling problem in IES with dynamically coupled multi-energy flows, focusing on the challenges posed by physical heterogeneity and PDE-constrained dynamics. A novel Many-objective Optimization Model, named MaOCMFM, is developed to represent the unsteady-state physical processes of electricity, gas, heat, and cooling subsystems using PDE formulations. To solve this complex problem efficiently, we proposed the PC-MaOEA-PINN algorithm, which integrates a PINN as a surrogate model into a probabilistic many-objective evolutionary framework. The use of cubic B-spline function encoding enables the algorithm to respect boundary conditions required for PDE learning, while guiding the search towards promising solution regions.

The proposed model and algorithm were validated on an IES test system constructed based on the 39-bus Matpower benchmark, coupled with realistic load, photovoltaic, and wind generation data derived from actual electricity market disclosures. A total of nine representative scenarios were designed to reflect realistic daily operation patterns. Experimental results demonstrated that PC-MaOEA-PINN consistently outperforms state-of-the-art many-objective algorithms in terms of conver-

gence, diversity, and computational efficiency, thereby confirming the practicality and superiority of the proposed approach.

In future work, we plan to improve PINNSM by incorporating meta-learning and multimodal data fusion techniques, aiming to better address the performance degradation of PINNs caused by variations in boundary conditions.

CRedit authorship contribution statement

Authorship Confirmation We, the undersigned, hereby confirm that: All authors listed on this manuscript have made significant contributions to the research and writing of this work. All authors have read and approved the final version of the manuscript. No individuals who qualify for authorship have been omitted. Originality and Exclusivity This manuscript is original and has not been published elsewhere, in whole or in part. This manuscript is not under consideration for publication elsewhere. The work does not contain any material that infringes upon existing copyrights, trademarks, patents, or any other intellectual property rights.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xingjuan Cai reports financial support was provided by the National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. System parameters

Table A.1

Parameters of power generation nodes in electronic subsystems.

Node	$P_{\max}$	$P_{\min}$	$Q_{\max}$	$Q_{\min}$	D^{CL}	U^{CL}	Type
30	1040	50	300	-300	500	500	CHP
31	646	50	50	-13	300	300	TH
32	725	70	300	-300	350	350	TH
33	652	60	200	-147	300	300	GT
34	508	50	120	-35	250	250	TH
35	687	60	30	-10	300	300	TH
36	580	50	50	-16	300	300	TH
37	564	50	24	-8	250	250	GT
38	-	-	-	-	400	400	pH
39	-	-	-	-	550	550	WIN

Table A.2

Parameters of heat source nodes in heating subsystem.

Node	$T_s^{\max}$	$T_s^{\min}$	$T_r^{\max}$	$T_r^{\min}$
1 (CHP)	120	110	80	60
2	120	110	80	60
3 (EB)	120	110	80	60
4	120	110	80	60
5	120	110	80	60
6	120	110	80	60

Table A.3

Parameters of heat source nodes in cooling subsystem.

Node	$T_s^{\max}$	$T_s^{\min}$	$T_r^{\max}$	$T_r^{\min}$
1 (ABS)	7	4	15	12
2	7	4	15	12
3 (ABS)	7	4	15	12
4	7	4	15	12
5	7	4	15	12
6	7	4	15	12

Table A.4

Parameters of pipelines in heating and cooling subsystem.

Branch	fbus	tbus	u	Length(m)	d	λ_w
1	1	2	957,060	2333.333	0.8	0.12
2	2	3	869,076	1166.667	0.8	0.12
3	3	4	516,888	1166.667	0.8	0.12
4	2	5	87,984	1166.667	0.8	0.12
5	3	6	352,188	1166.667	0.8	0.12

Table A.5

Parameters of pipelines in gas subsystem.

Branch	fbus	tbus	λ	Branch	fbus	tbus	λ
1	1	2	0.07	11	11	12	0.86
2	2	3	0.40	12	12	13	0.91
3	3	4	0.39	13	13	14	0.26
4	5	6	0.10	14	14	15	0.63
5	6	7	0.15	15	15	16	0.45
6	7	4	0.22	16	11	17	0.05
7	4	14	0.66	17	17	18	0.006
8	8	9	0.26	18	18	19	0.002
9	9	10	0.81	19	19	20	0.03
10	10	11	0.45	–	–	–	–

Table A.6

Parameters of each node in gas subsystem.

Node	$p^{\max}$	$p^{\min}$	$v^{\max}$	$v^{\min}$	Node	$p^{\max}$	$p^{\min}$
1 (GS)	77	10	1.7931	0.9	11	66.2	10
2 (GS)	77	10	1.26	0	12	66.2	10
3 (CHP)	80	30	–	–	4	80	10
13(GS)	66.2	10	0.27	0	7	80	30
5 (GS)	77	10	0.72	0	15	66.2	10
6 (GT)	80	30	–	–	16	66.2	10
14(GS)	66.2	10	1.44	0	17	66.2	10
8 (GS)	66.2	10	2.3018	1	18	80	10
9	66.2	10	–	–	19 (GT)	66.2	10
10	66.2	30	–	–	20	66.2	25

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